

Schottky Diode – 25/11/2022

Thursday, 11 May 2023 15:10

Setup:

- Dip probe with attached diode and Pt_1000 – Probe to allow the measurements near/ in liquid nitrogen for the experiment. Pt_1000 resistance thermometer used for high accuracy. Diode being tested for experiment.
- Multimeter – reads and displays the current and voltage from diode and Pt_1000 as well as the resistance.
- Keithley source generator – takes sweeps a range of voltages between a selected start and end voltage and displays data.
- Dewar containing liquid nitrogen – needed to take the diode to a low enough temperature to study the e

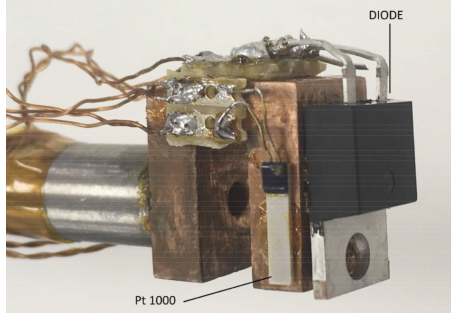


Figure 1. End of dip probe containing the diode and Pt_1000.

Task 1:

The first task is to identify which wires in the break out cable are assigned to which part of the head of the dip probe:

This was done by measuring the resistance, using a hand held multimeter, between the copper tracks at the end of the dip probe and the metal part of the banana plugs. When the multimeter beeped it indicated which combinations were connected. These were noted down for use when connecting the banana plugs to the multimeter and source generator.

Task 2:

The second task was to solder the diode and Pt_1000 to the copper tracks identified in task 1 at the end of the dip probe:

Making sure to insert diode into the end hole and soldering each one of the two legs to the two adjacent tracks as in figure 3. As well as this making sure to place the Pt_1000 along the side of the end of the dip probe.

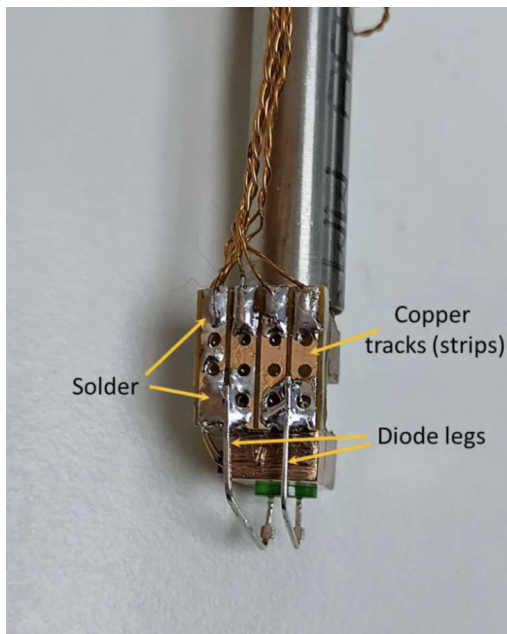


Figure 3. Soldering diode to 4-track board.



Rule for

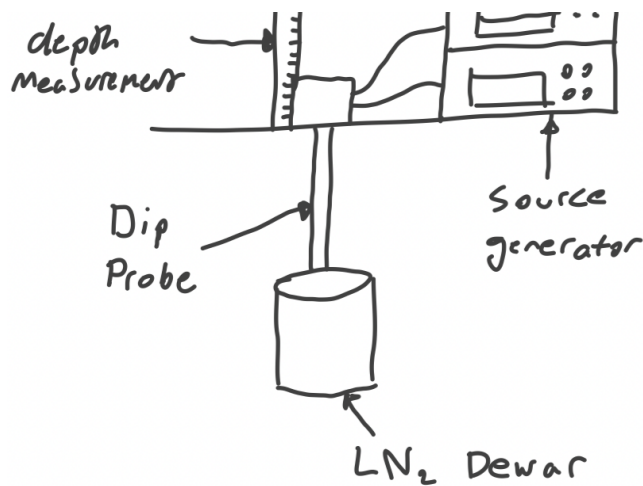


Figure 2. Diagram of the experiment set up.

TM

Task 3:

Task 3 was to identify which way round the voltage was to be swept:

This was done by performing an IV sweep across a suitable current range not exceeding 1A to measure the relation at room temperature for both directions. Following the instructions in the lab script on how to set up the Keithley source generator and perform an IV sweep.

Room temperature sweep			
Reading	Unit	Value	Unit
1.25E-09	Amp DC	-0.0003053	Volt DC
1.62E-07	Amp DC	0.04968889	Volt DC
1.27E-06	Amp DC	0.09979274	Volt DC
8.63E-06	Amp DC	0.14960954	Volt DC
6.09E-05	Amp DC	0.19974458	Volt DC
0.00041197	Amp DC	0.24950577	Volt DC
0.00278326	Amp DC	0.29958326	Volt DC
0.01807609	Amp DC	0.34970343	Volt DC
0.09504796	Amp DC	0.39949322	Volt DC
0.36820054	Amp DC	0.44985679	Volt DC
0.85603291	Amp DC	0.49984467	Volt DC
1.00001919	Amp DC	0.51166368	Volt DC

Table 1. Room temperature IV sweep current and voltage readings.

Task 4:

Task 4 was to identify a voltage range needed to provide good data. And also select appropriate points to measure:

The decision was made to start the voltage at 0V and stop at 1.8V with steps of 150mV. Therefore giving 12 counts. There was a total of 7 sweeps including the room temperature sweep. Careful measurements of the depth of the liquid nitrogen in the dewar and the height of the dip probe. Markings using a black marker were made on the dip probe at 7 points to indicate where to lower to. These points and number of them were picked as to scan fast enough to avoid thermal drift but enough data points to provide good data to produce a nice curve fit when fitted next to the exponential equation, $I = I_0 \left(\exp\left(\frac{eV}{KT}\right) - 1 \right)$.

Task 5:

Task 5 was to make IV sweeps at a range of temperatures:

The change in temperature was controlled by the lowering of the dip probe in to the liquid nitrogen dewar. The markings made in task 4 were the indicator of the level needed to be lowered to each time. After lowering the dip probe it was curtail to let the system stabilize before performing IV sweeps to ensure good data. Each sweep was saved separately onto a USB stick to allow good plotting later on. And to also allow each sweep data set to be safely saved before executing another sweep. Taking note of the resistance on the Pt_1000 at each sweep was crucial to later go and calculate the exact temperatures.

Room temperature sweep, 1.089162		Measurement 2, 0.9612062		Measurement 3, 0.8292062		Measurement 4, 3.52	
Reading	Unit	Reading	Unit	Reading	Unit	Reading	Unit
1.25E-09	Amp DC	2.30E-09	Amp DC	3.04E-10	Amp DC	1.82E-10	Amp DC
1.62E-07	Amp DC	5.27E-09	Amp DC	1.54E-10	Amp DC	5.87E-11	Amp DC
1.27E-06	Amp DC	2.22E-08	Amp DC	2.25E-10	Amp DC	-1.29E-10	Amp DC
8.63E-06	Amp DC	7.83E-08	Amp DC	1.87E-09	Amp DC	3.16E-11	Amp DC
6.09E-05	Amp DC	1.19E-07	Amp DC	1.50E-08	Amp DC	5.30E-11	Amp DC
0.00041197	Amp DC	7.11E-06	Amp DC	7.44E-08	Amp DC	3.81E-09	Amp DC
0.00278326	Amp DC	6.78E-06	Amp DC	9.25E-07	Amp DC	7.87E-09	Amp DC
0.01807609	Amp DC	0.00063937	Amp DC	1.25E-06	Amp DC	4.10E-08	Amp DC
0.09504796	Amp DC	0.39949322	Amp DC	0.00018235	Amp DC	9.29E-07	Amp DC
0.36820054	Amp DC	0.44985679	Amp DC	0.00246223	Amp DC	2.48E-05	Amp DC
0.85603291	Amp DC	0.49984467	Amp DC	0.03181052	Amp DC	0.00094655	Amp DC
1.00001919	Amp DC	0.51166368	Amp DC	0.49975469	Amp DC	0.49970904	Amp DC

1.000019190 Amp DC	0.511603675 Volt DC	0.308020876 Amp DC	0.548671113 Volt DC	0.289997753 Amp DC	0.548903982 Volt DC	0.013870662 Amp DC	0.549675107 Volt DC
1.000004464 Amp DC	0.511603675 Volt DC	1.000004464 Amp DC	0.550504763 Volt DC	0.999979489 Amp DC	0.550504763 Volt DC	0.999979489 Amp DC	0.550504763 Volt DC
Measurement 5, 1.6240							
Reading	Unit	Value	Unit	Reading	Unit	Value	Unit
-8.23E-11	Amp DC	-0.000000000	Volt DC	-1.87E-11	Amp DC	-0.000000000	Volt DC
-1.03E-10	Amp DC	0.048057407	Volt DC	0.96E-11	Amp DC	0.048057407	Volt DC
-2.03E-10	Amp DC	0.099032004	Volt DC	4.59E-10	Amp DC	0.099032004	Volt DC
-2.03E-10	Amp DC	0.148030111	Volt DC	-1.80E-10	Amp DC	0.148030111	Volt DC
2.03E-10	Amp DC	0.199030060	Volt DC	3.11E-10	Amp DC	0.199030060	Volt DC
-8.00E-11	Amp DC	0.249030423	Volt DC	-2.60E-10	Amp DC	0.249030423	Volt DC
2.30E-10	Amp DC	0.299029978	Volt DC	3.50E-10	Amp DC	0.299029978	Volt DC
2.59E-09	Amp DC	0.34977302	Volt DC	-2.59E-10	Amp DC	0.34977302	Volt DC
4.07E-09	Amp DC	0.3994538	Volt DC	4.74E-09	Amp DC	0.3994538	Volt DC
5.43E-07	Amp DC	0.44903446	Volt DC	5.46E-08	Amp DC	0.44903446	Volt DC
2.19E-06	Amp DC	0.498657903	Volt DC	1.35E-06	Amp DC	0.498657903	Volt DC
0.200400979	Amp DC	0.54890604	Volt DC	7.59E-05	Amp DC	0.54890604	Volt DC
0.00016947	Amp DC	0.599600819	Volt DC	0.00444418	Amp DC	0.599600819	Volt DC
0.59900983	Amp DC	0.648774373	Volt DC	0.190334615	Amp DC	0.648774373	Volt DC
1.000001073	Amp DC	0.691903918	Volt DC	1.000006293	Amp DC	0.691903918	Volt DC

Table 2. All IV sweeps at different temperatures.

Task 6:

Task 6 was to calculate the temperature that each sweep was performed at:

Calibration of the resistance measurements would allow the extraction of the temperatures.

Using the equation, $R(T) = R(0)(1 + \alpha T)$, (with T in degrees Celsius).

The first step was to find value of α .

The resistance at room temperature was correct at $1085.1\Omega \pm 0.1\Omega$. Which is in the region of 1050Ω , which was expected. It could also be assumed that there is a temperature of 77K when the resistor is submerged. And a resistance of 1000Ω when at 0°C . Also the assumption of room temperature was the universally standard 20°C .

$$R(0) = 1000\Omega$$

$$1085.1\Omega \pm 0.1\Omega = 1000\Omega(1 + \alpha 20^\circ\text{C})$$

$$(1085.1\Omega \pm 9.216 \times 10^{-3}\%) \div 1000\Omega = 1 + \alpha 20^\circ\text{C}$$

$$1.0851\Omega \pm 0.01\Omega = 1 + \alpha 20^\circ\text{C}$$

$$0.0851\Omega \pm 0.01\Omega = \alpha 20^\circ\text{C}$$

$$4.255 \times 10^{-3} K^{-1} \pm 0.01 K^{-1} = \alpha$$

$$4.255 \times 10^{-3} K^{-1} \pm 0.01 K^{-1} = \alpha$$

$$\text{The accepted value of } \alpha = 3.925 \times 10^{-3} K^{-1}$$

$$\text{Therefore, } \Delta\alpha = 3.3 \times 10^{-4} K^{-1}$$

This makes sense as this calculated value for α is within the uncertainty.

The reason the use of resistance thermometers is implemented is because it measures to a high level of accuracy. It also has good linear characteristics over a wide range of temperatures.

Task 7:

Task 7 was to plot each of the IV curves on a single graph and also determine the height of the Schottky barrier as a function of temperature and plot:

To plot the IV curves, the collected data from the IV sweeps were inputted into python and plotted.

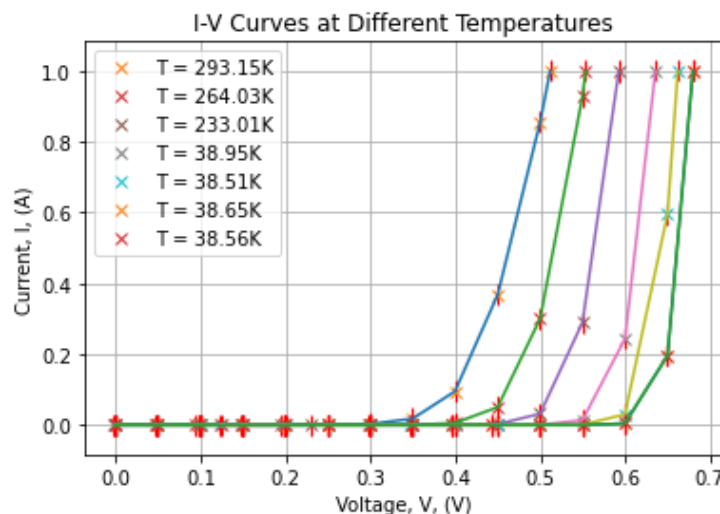


Figure 4. I-V curves at different temperatures.

For the barrier height:

The saturation current is initially the first value to identify using the equation,

$I = I_0 \left(\exp\left(\frac{eV}{KT}\right) - 1 \right)$, I the current, I_0 the saturation current, e the charge of an electron, V the voltage, K the Boltzmann constant, T the temperature.

Once the saturation current was determined. It could then be entered into $I_0 = AA^*T^2 \exp\left(-\frac{e\phi_{bn}}{K_B T}\right)$, where A is surface area of contact between metal and semiconductor materials inside the diode ($3mm^2$), A^* is the Richardson constant ([Equation]), [Equation] is Schottky barrier height.

Reading, (Kelvin, K)	Barrier Height	Units
Room Temperature, 293.15	0.8504040597	eV
264.031316	0.8536257704	eV
233.008989	0.8524536427	eV
38.9549354	0.6670247135	eV
38.5140423	0.6929338201	eV
38.6494125	0.7108610688	eV
38.5554054	0.7107441763	eV

Table 3. Barrier height at different temperatures.

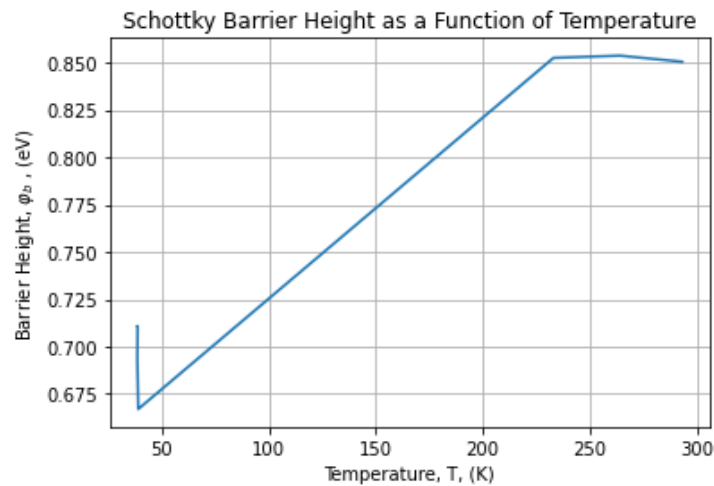


Figure 5. Schottky barrier height as a function of temperature.

Task 8:

Task 8 was to discuss the temperature dependence observed for the barrier height and explain it in terms of physics responsible for the electron transport across the metal-semiconductor interface:

The observed and plotted temperature dependence of the barrier height is that as temperature increases so does barrier height. And as temperature decreases so does barrier height. The plot looks very linear with exceptions at the extremes. However, this is due to the sudden and large drop in temperature in the data causing the line fit jump causing the linear plot. The increase in barrier height at the lowest temperature was unexpected. After thought and discussion thought it may be due to evaporation of the liquid nitrogen around the probe causing unstable temperatures.

This makes sense and follows the relevant literature on this topic. Even with the data not being as accurate at the extremes of the temperature scale. I researched and read some of the literature to further my understanding. Figure 6 also follows the linear trend like my own plot in figure 5.

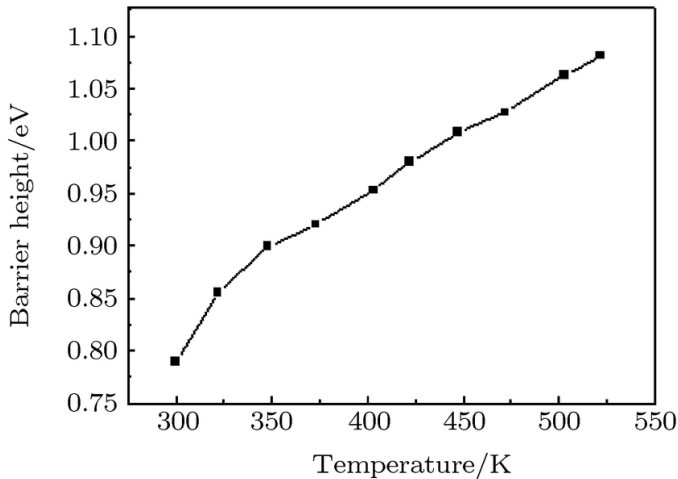


Figure 6. Plot of barrier height as a function of temperature.
From:
<https://iopscience.iop.org/article/10.1088/1674-1056/20/6/067303/pdf>

In Schottky diode there are three mechanisms by which electrons are carried across the barrier: thermionic emission, tunnelling through the barrier and carrier recombination. In this temperature range thermionic emission is the main mechanism for transport through the diode. This is where the thermal energy of the charge carrier over comes the work function. As the barrier height decreases due to colder temperatures so too does the work function. If the work function is smaller than that of the charge carriers then thermionic emission takes place across the metal-semiconductor interface.

Python Scripts:

Figure 4. I-V curves at different temperatures:

```
import numpy as np
import matplotlib.pyplot as plt

#Data for room temp
I_roomtemp = np.array([1.25e-09,1.62e-07,1.27e-06,8.63e-06,6.09e-05,0.000411969,0.002783258,0.01807609,0.095047958,0.368200541,0.856032908,1.000019193])
V_roomtemp = np.array([-0.000305304,0.049688887,0.099792741,0.149609536,0.199744582,0.249505773,0.299583256,0.349703431,0.399493217,0.449856788,0.49984467,0.511663675])

#Data for reading 2
I_2 = np.array([2.39e-09,5.27e-09,2.22e-08,7.83e-08,7.60e-07,7.11e-06,6.76e-05,0.000639397,0.005914759,0.050420664,0.300701678,0.928252876,1.000008464])
V_2 = np.array([-0.000597315,0.046884462,0.093531713,0.125425696,0.19971332,0.249424711,0.299458981,0.349569559,0.399617821,0.449712336,0.499863505,0.549671113,0.553504705])

#Data for reading 3
I_3 = np.array([3.04e-10,1.54e-10,2.25e-10,1.67e-09,1.56e-08,7.44e-08,9.25e-07,1.32e-05,0.000182315,0.002487237,0.031810582,0.289999753,0.998976469])
V_3 = np.array([-0.00072196,0.049720995,0.099469222,0.148179889,0.194074377,0.229658589,0.299409986,0.349717021,0.39972347,0.44980402,0.499775469,0.549693882,0.592406511])

#Data for reading 4
I_4 = np.array([1.52e-10,5.97e-11,-1.29e-10,3.16e-11,5.30e-11,3.91e-09,7.97e-09,4.10e-08,9.29e-07,2.48e-05,0.00059455,0.013878662,0.243763402,0.999927104])
V_4 = np.array([-0.001877224,0.048462916,0.098498352,0.148613438,0.199939206,0.123216607,0.296754271,0.349770784,0.399512738,0.44980976,0.499709904,0.549675107,0.599746466,0.635563374])

#Data for reading 5
I_5 = np.array([8.22e-11,-1.03e-10,2.03e-10,-2.02e-10,2.92e-10,-8.00e-11,2.33e-10,2.98e-09,4.07e-08,5.43e-07,2.18e-05,0.000818079,0.03016947,0.595009983,1.000001073])
V_5 = np.array([-0.000837372,0.048257407,0.099832304,0.14885211,0.198938206,0.249638423,0.29882881,0.347379953,0.39474839,0.449736446,0.499822557,0.54946059,0.599660218,0.649741828,0.661903918])

#Data for reading 6
I_6 = np.array([1.97e-11,9.99e-11,4.59e-10,-1.93e-10,3.11e-10,-2.60e-10,3.50e-10,-2.29e-10,4.74e-09,5.46e-08,1.35e-06,7.69e-05,0.00444418,0.196334615,1.000058293])
V_6 = np.array([-0.00088469,0.048935954,0.099050865,0.149391964,0.199362665,0.249165997,0.299289078,0.34977302,0.3964338,0.4420485
```

```

5,0.499857903,0.549488604,0.599620819,0.649774373,0.679699123]
)

#Data for reading 7
I_7 = np.array([1.97e-11,9.99e-11,4.59e-10,-1.93e-10,3.11e-
10,-2.60e-10,3.50e-10,-2.29e-10,4.74e-09,5.46e-08,1.35e-
06,7.69e-05,0.00444418,0.196334615,1.000058293])
V_7 =
np.array([-0.00088469,0.048935954,0.099050865,0.149391964,0.19
9362665,0.249165997,0.299289078,0.34977302,0.3964338,0.4420485
5,0.499857903,0.549488604,0.599620819,0.649774373,0.679699123]
)

#Plotting
plt.xlabel('Voltage, V, (V)')
plt.ylabel('Current, I, (A)')
plt.title('I-V Curves at Different Temperatures')
plt.grid()

plt.errorbar(V_roomtemp, I_roomtemp, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_roomtemp,I_roomtemp,'x', label='T = 293.15K')

plt.errorbar(V_2, I_2, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_2,I_2,'x', label='T = 264.03K')

plt.errorbar(V_3, I_3, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_3,I_3,'x', label='T = 233.01K')

plt.errorbar(V_4, I_4, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_4,I_4,'x', label='T = 38.95K')

plt.errorbar(V_5, I_5, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_5,I_5,'x', label='T = 38.51K')

plt.errorbar(V_6, I_6, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_6,I_6,'x', label='T = 38.65K')

plt.errorbar(V_7, I_7, yerr=0.03, ecolor='red',
elinewidth=1.2)
plt.plot(V_7,I_7,'x', label='T = 38.56K')

plt.legend()
plt.show()

```

Figure 5. Schottky barrier height as a function of temperature

```

import numpy as np
import matplotlib.pyplot as plt
import math

#Data
barrier_height =
np.array([0.8504040597,0.8536257704,0.8524536427,0.6670247135,
0.6929338201,0.7108610688,0.7107441763])

temprature =
np.array([293.15,264.031316099,233.00898942,38.9549354,38.5140
423,38.6494125,38.5554054])

```

```
#Plotting
plt.xlabel('Temperature, T, (K)')
plt.ylabel('Barrier Height,  $\phi_b$ , (eV)')
plt.title('Schottky Barrier Height as a Function of
Temperature')
plt.grid()

plt.plot(temperature, barrier_height)
```

MOSFET - 21/3/23

Tuesday, April 04, 2023 11:54 PM

Setup:

1. Keithley power supply – digital DC power supply capable of safely supplying 30V. For this experiment it is set to 10V at 0.2A.
2. USB-6009 Data Acquisition (DAQ) card – controlled with LabView. Provides basic functionality with 8 analog inputs, 2 analog outputs and 12 I/O digital channels. This is used to connect the circuit to LabView to analyze and control the circuit.
3. Circuit to characterize the MOSFET transistor – made on a bread board. Circuit shown below.

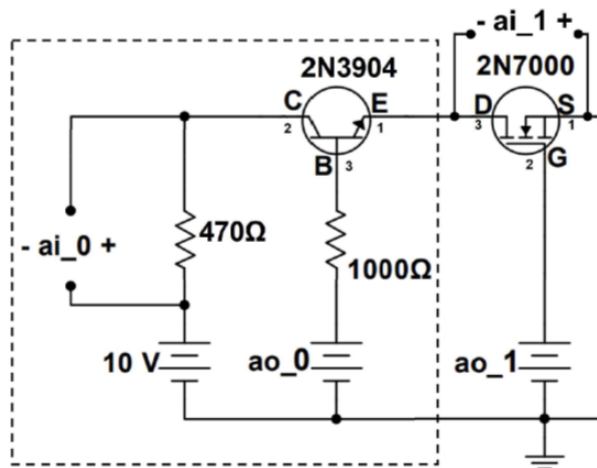


Figure (1): Circuit to characterize the MOSFET transistor.

4. LabView – MOSFET analysis routine set up to allow the analysis of the field-effect behavior. Routine shown below.

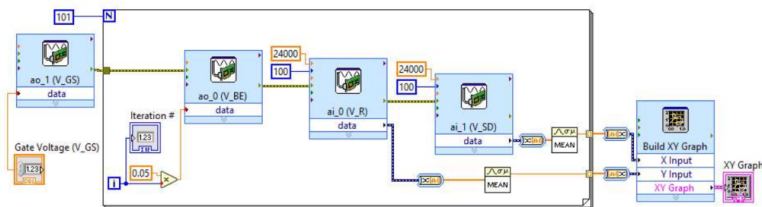


Figure (2): MOSFET analysis routine.

The general set up of the experiment is shown below. The DAQ card is connected to a computer for analysis and control in LabView.

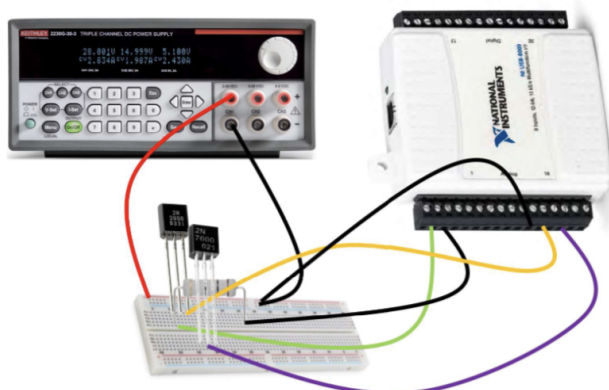


Figure (3): Schematic of MOSFET curcuit setup.

Task 1:

Find the threshold voltage using constant current method:
Setting ao0 channel to a high value and slowly increasing the voltage on the ao1 channel using DAQ_Control VI, until a movement in ai_1, it was found that $V_{thr} = 1.267V$.

Task 2:

Acquire appropriate I-V measurements of the transistor at applied gate voltage, V_{GS} .
Measurement's were taken below V_{thr} to show the significance of threshold voltage. I.e. there is no defined saturation and linear region.

Analysis task 1:

Plot of i_D against V_{SD} at varying V_{GS} .

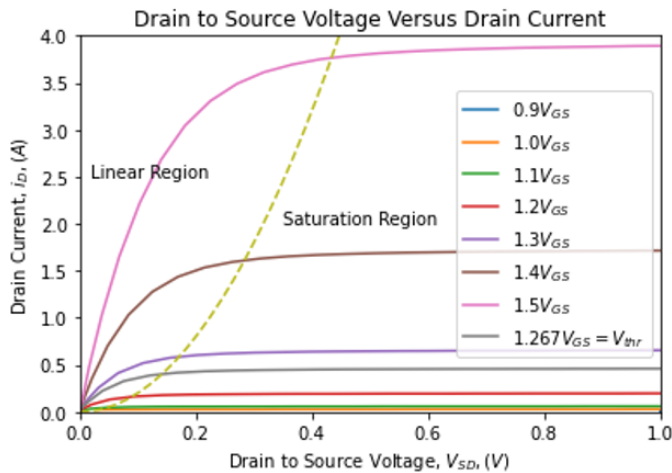


Figure (4): Drain to Source Voltage versus Drain Current at varying Gate to Source Voltages.

Data followed the expected pattern. A clearly defined linear region and saturation region. Shown to be separated by an x^2 curve, shown as the dotted line. The $i_D - V_{SD}$ characteristics were highly dependent on V_{GS} .

Analysis task 2:

Employ the linear model and fit the curve over the graph.

Linear model equation:

$$i_D = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{thr}) V_{DS}, \text{ for } |V_{DS}| \ll (V_{GS} - V_{thr})$$

(Where μ is field effect mobility, $C_{ox} = 10 \pm 1 \mu F cm^{-2}$, $W = 100 \pm 5 \mu m$, $L = 5 \pm 0.1 \mu m$, $V_{thr} = 1.267V$)

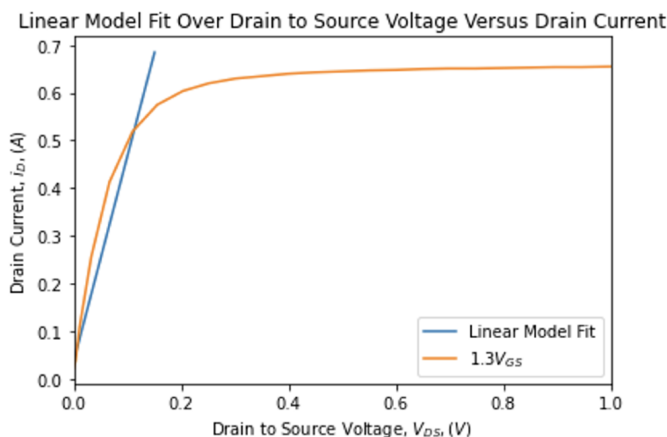


Figure (5): Linear model fit over drain to source voltage versus drain current.

The linear model does not completely fit the linear region of the curve exactly. The combined error and therefore the maximum deviation from the linear model for the data in figure 3.2 is $\pm 0.0225 \text{ A/V}$. This is not an insignificant value and a new aim for the experiment should be to try and reduce this error. However, it is small enough for the model to still fit the data well enough to draw conclusions about the behavior of the field-effect in MOSFET's.

Potential sources of inaccuracy in the data:

Temperature effects: MOSFET characteristics are temperature-dependent, and variations in temperature during the experiment can affect the device's behavior. Changes in temperature can lead to variations in threshold voltage, carrier mobility, and other parameters, introducing uncertainties in the measurements. - Next time conducting the experiment in a temperature controlled environment where, and wearing PPE that stops body heat interacting with the setup.

Measurement setup and connections: Improper connections, loose connections, or inadequate shielding can introduce noise and errors in the measurement circuit. Proper care should be taken in the experimental setup to ensure accurate and reliable measurements. - Next time the use of gloves when setting up the circuit would help reduce the inaccuracies as well as using checked equipment.

Codes for Plots:

Figure (4) code:

```
#imports
import numpy as np
import matplotlib.pyplot as plt
import pandas as pa
#V=0.9
V_09 = np.loadtxt('mosfet voltage is 0.9V.txt',
skiprows=1)
V_09_V_SD = V_09[:, 0]
V_09_i_D = V_09[:, 1]

#V=1
V_1 = np.loadtxt('mosfet voltage is 1V.txt',
skiprows=1)
V_1_V_SD = V_1[:, 0]
V_1_i_D = V_1[:, 1]

#V=1.1
V_11 = np.loadtxt('mosfet voltage is 1.1V.txt',
skiprows=1)
V_11_V_SD = V_11[:, 0]
V_11_i_D = V_11[:, 1]

#V=1.2
V_12 = np.loadtxt('mosfet voltage is 1.2V.txt',
skiprows=1)
V_12_V_SD = V_12[:, 0]
V_12_i_D = V_12[:, 1]
```



```

#V=1.3
V_13 = np.loadtxt('voltage is 1.3V MOSFET.txt',
skiprows=1)
V_13_V_SD = V_13[:, 0]
V_13_i_D = V_13[:,1]

#V=1.4
V_14 = np.loadtxt('mosfet voltage is 1.4.txt',
skiprows=1)
V_14_V_SD = V_14[:, 0]
V_14_i_D = V_14[:,1]

#V=1.5
V_15 = np.loadtxt('mosfet voltage is 1.5V.txt',
skiprows=1)
V_15_V_SD = V_15[:, 0]
V_15_i_D = V_15[:,1]

#threshold data (V=1.267)
V_threshold_1267 = np.loadtxt('threshohd voltage
is 1.267V MOSFET.txt', skiprows=1)
V_threshold_V_SD = V_threshold_1267[:, 0]
V_threshold_i_D = V_threshold_1267[:, 1]

x = np.linspace(0,1)
y = 20*x**2

#plotting
plt.plot(V_09_V_SD, V_09_i_D,
label='$0.9V_{GS}$')
plt.plot(V_1_V_SD, V_1_i_D,
label='$1.0V_{GS}$')
plt.plot(V_11_V_SD, V_11_i_D,
label='$1.1V_{GS}$')
plt.plot(V_12_V_SD, V_12_i_D,
label='$1.2V_{GS}$')
plt.plot(V_13_V_SD, V_13_i_D,
label='$1.3V_{GS}$')
plt.plot(V_14_V_SD, V_14_i_D,
label='$1.4V_{GS}$')
plt.plot(V_15_V_SD, V_15_i_D,
label='$1.5V_{GS}$')
plt.plot(V_threshold_V_SD, V_threshold_i_D,
label='$1.267V_{GS}=V_{thr}$')
plt.plot(x,y, '--')
plt.annotate('Saturation Region', (0.35,2))
plt.annotate('Linear Region', (0.02,2.5))

plt.legend()
plt.xlabel('Drain to Source Voltage, $V_{SD}$,
(V)$')
plt.ylabel('Drain Current, $i_D$, (A)$')
plt.title('Drain to Source Voltage Versus Drain
Current')

```

```
plt.xlim(0,1)
plt.ylim(0,4)
```

Figure (5) code:

```
import numpy as np
import matplotlib.pyplot as plt

#V=1.3
V_13 = np.loadtxt('voltage is 1.3V MOSFET.txt',
skiprows=1)
V_13_V_SD = V_13[:, 0]
V_13_i_D = V_13[:,1]

x = np.linspace(0,0.15)

#constants
L = 5*10**-6
W = 100*10**-6
C_ox = 10*10^-10
V_thr = 1.267
V_13_V_GS = 1.3

m = 4.3
c = 0.04
y = m*x+c

plt.plot(x, y, label='Linear Model Fit')

#plotting
plt.plot(V_13_V_SD, V_13_i_D,
label='$1.3V_{GS}$')
plt.legend()
plt.xlabel('Drain to Source Voltage, $V_{DS}$, (V)$')
plt.ylabel('Drain Current, $i_D$, (A)$')
plt.title('Linear Model Fit Over Drain to Source Voltage Versus Drain Current')
plt.xlim(0,1)
```

Superconductivity – 18/11/2022

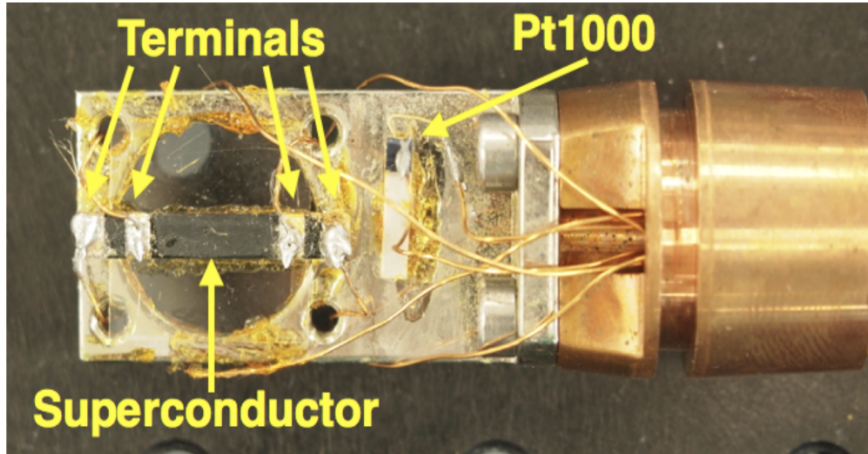
Thursday, May 11, 2023 5:12 PM

Task 1:

First to prepare the probe, identify all the connections and wire each contact up to the instrumentation correctly.

The superconductor and a Pt1000 thermometer are pre-wired and mounted on the cryostat probe. Connect the probe to the blue break out box with the cable. Measuring the resistance of both the Pt1000 and the superconductor and note these values with a measurement of the room temperature.

Below is a figure for the end of the probe used.



Task 2:

Pump the cryostat sample space down to a low pressure in preparation for making finely controlled temperature changes then cool it to 77K using liquid nitrogen.

1. Close the valve on the vacuum connection for the outer jacket.
2. Stop the turbo pump. now wait 15 minutes whilst the turbo pump blades slow down.
3. Allow the turbo pump to slow down, carefully unscrew the wing nut on the clamping ring to release the clamping ring.
4. Connect the vacuum hose to the sample space flange. Make sure the flanges of the hose and the sample space pipe are aligned. Open the valve connected to the sample space and start the turbo-pump again. It will take about 20 minutes for the sample space to reach the required pressure.

Task 3:

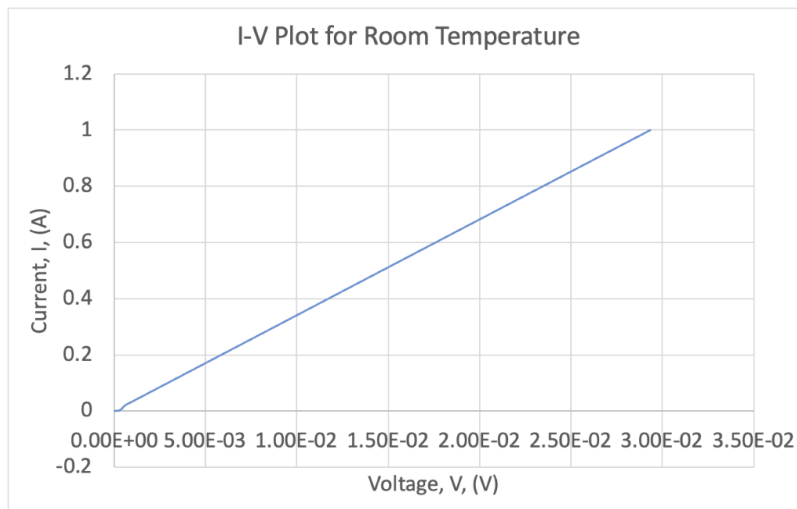
Take V-I measurements every 10 K to identify a rough temperature range in which the transition occur so a more careful investigation can be performed in smaller steps so you have good data covering the full transition region.

Task 4:

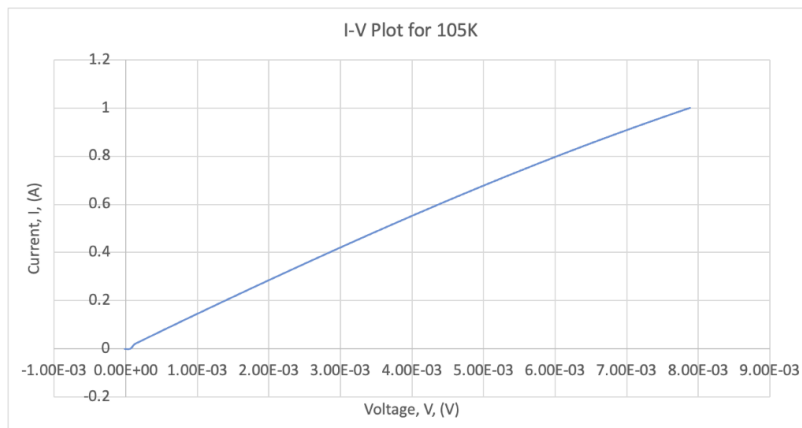
Now obtain a measurement of the transition by taking many data points over a small temperature range taken in steps of at least every 1 K between the lower and upper boundaries of the transition temperature, so you can quantify the phase change into the superconducting state. Transition temperature was taken and noted during the warming and cooling to identify any hysteresis in the measurements.

Analysis Task 1:

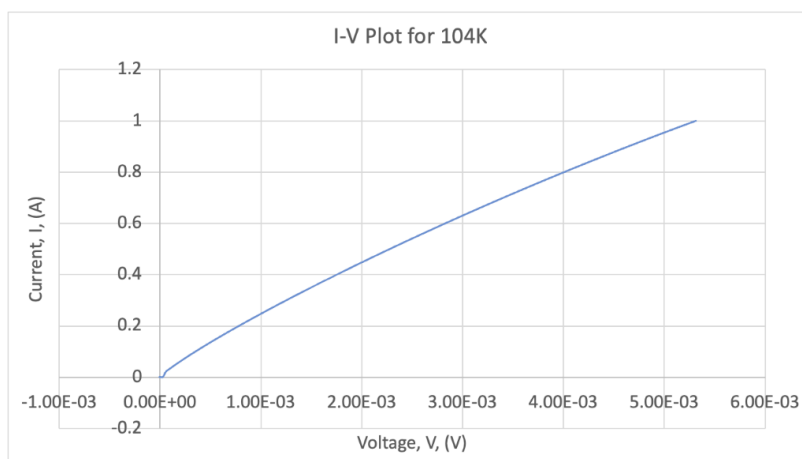
Determining the resistance of the superconductor at each temperature. And checking if the V-I plots show ohmic behaviour.



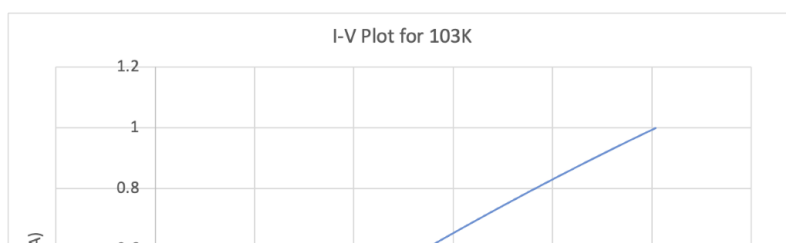
The resistance for Room Temp would be 0.03Ω and shows very strong ohmic behavior being a straightline from the origin.

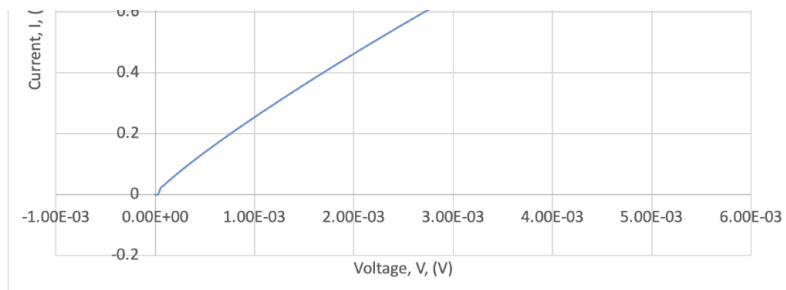


The resistance for 105K would be $8 \times 10^{-3}\Omega$ and shows very strong ohmic behavior being a Straightline from the origin.

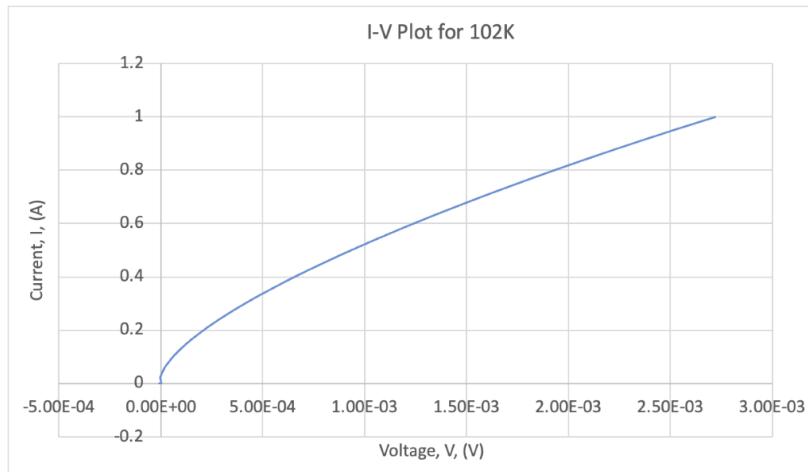


The resistance for 104K would be 5.3×10^{-3} and shows strong ohmic behavior being a straightline from the origin.

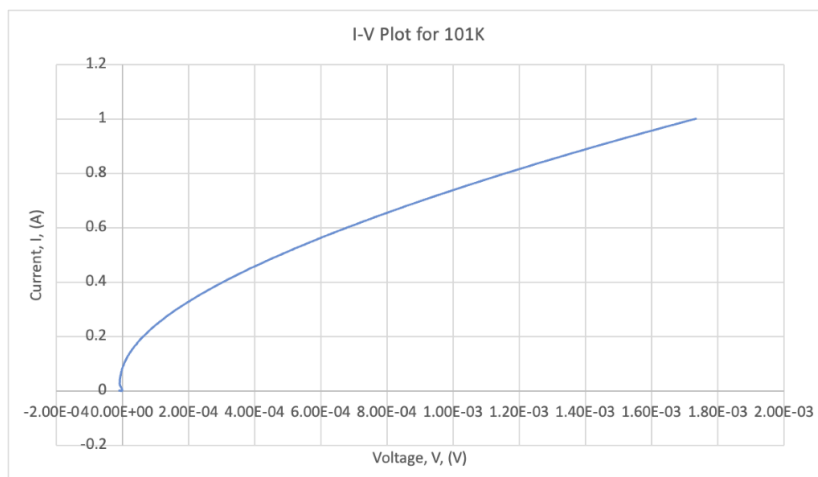




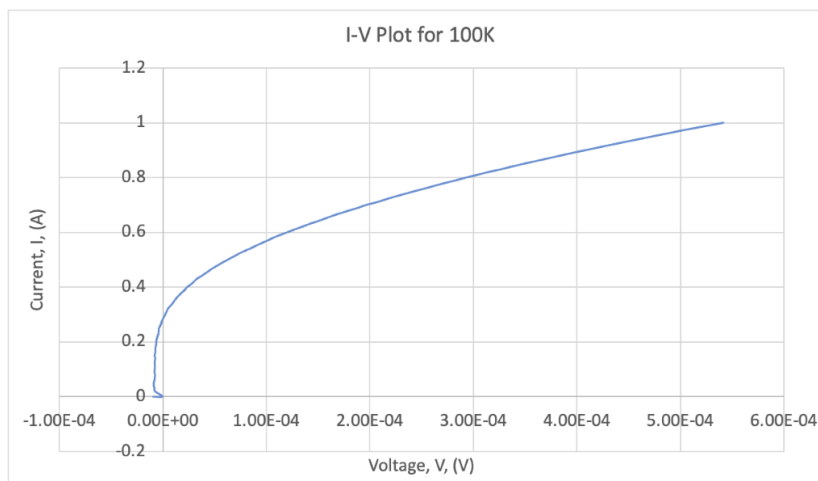
The resistance for 103K would be $5 \times 10^{-3} \Omega$ and still shows ohmic behavior however is starting to curve.



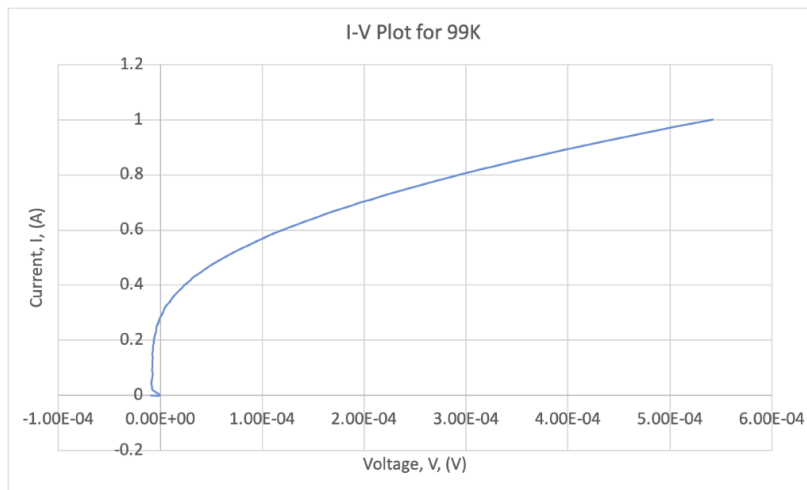
The resistance for 102K would be $3.1 \times 10^{-3} \Omega$. the ohmic behavior is breaking down at this point near the origin.



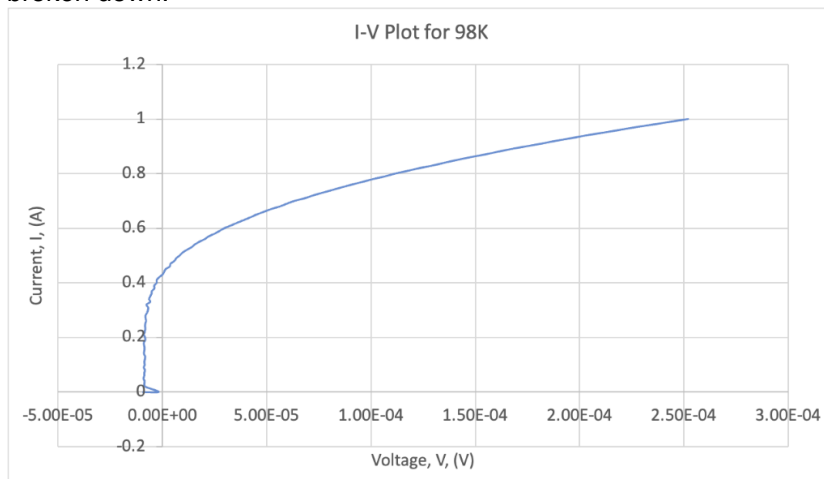
The resistance for 101K is $2.9 \times 10^{-3} \Omega$. The ohmic behavior is beginning to break down.



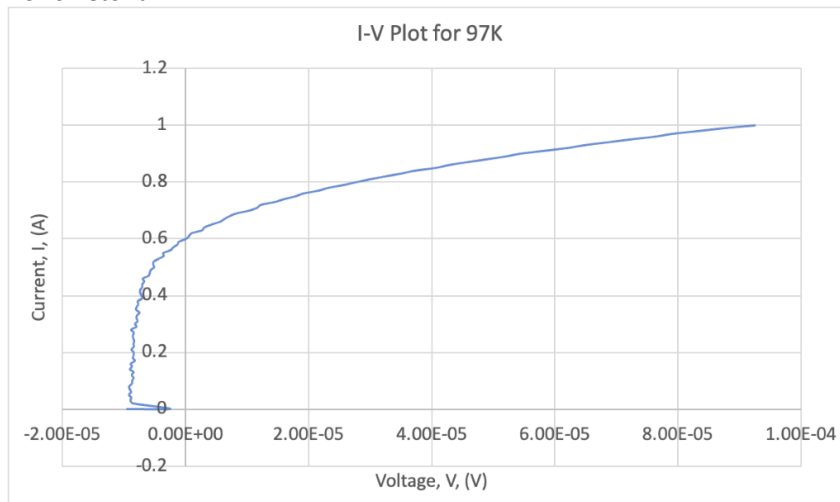
The resistance for 100K is $1.125 \times 10^{-3} \Omega$. The ohmic behavior can only be seen after $1 \times 10^{-4} \text{V}$.



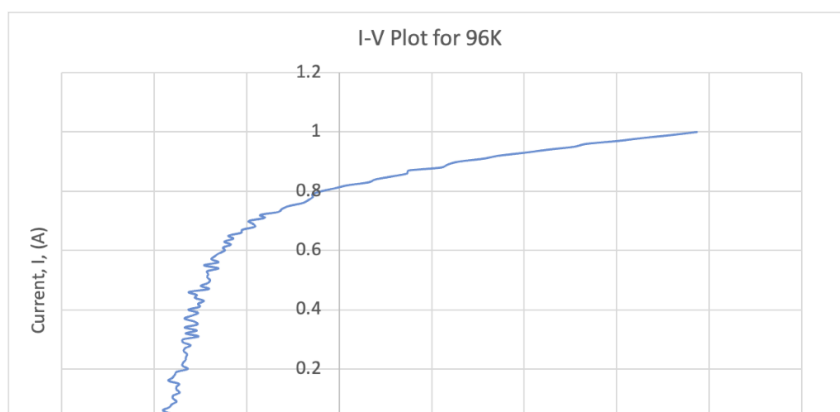
The resistance for 99K is $1 \times 10^{-3} \Omega$. Ohmic behavior has nearly completely broken down.

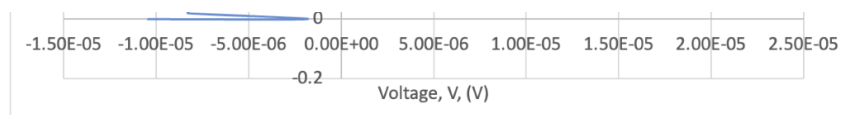


The resistance for 98K is $2.5 \times 10^{-4} \Omega$. The gradient at low voltages is nonexistent.

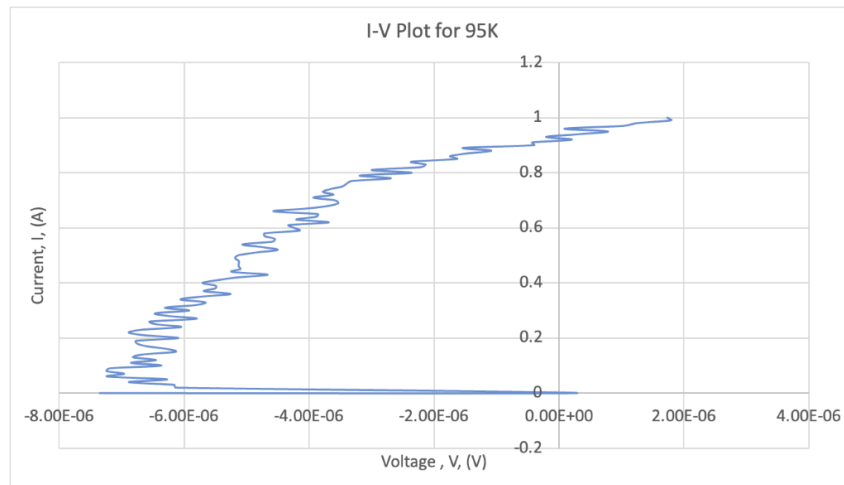


The resistance for 97K is around $1 \times 10^{-4} \Omega$. Data is getting unstable around low voltages.

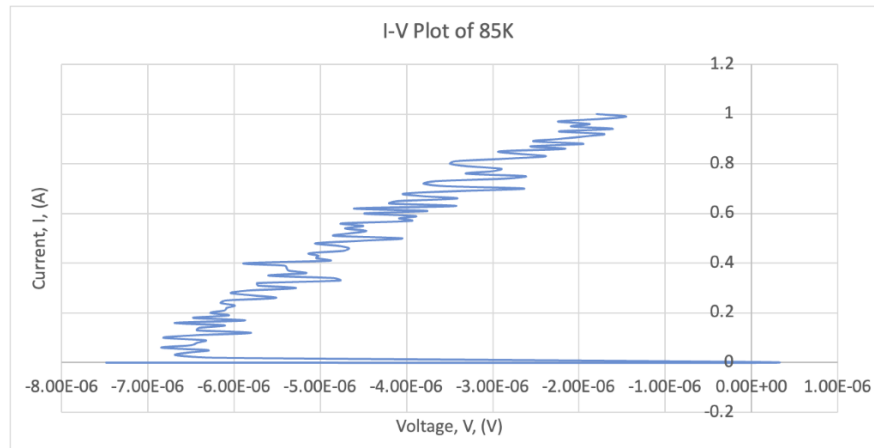




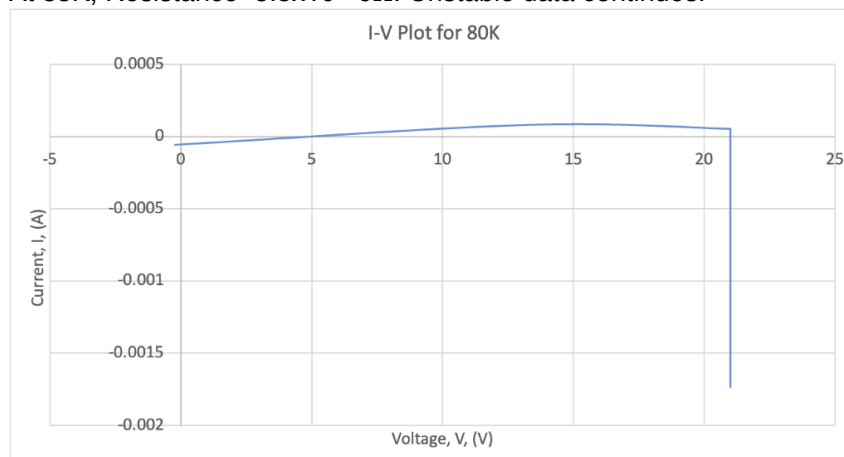
The resistance for 96K is around $3 \times 10^{-5} \Omega$. All ohmic behavior has broken down. The data at low V is also very unstable. Potentially due to liquid nitrogen evaporation around the tip.



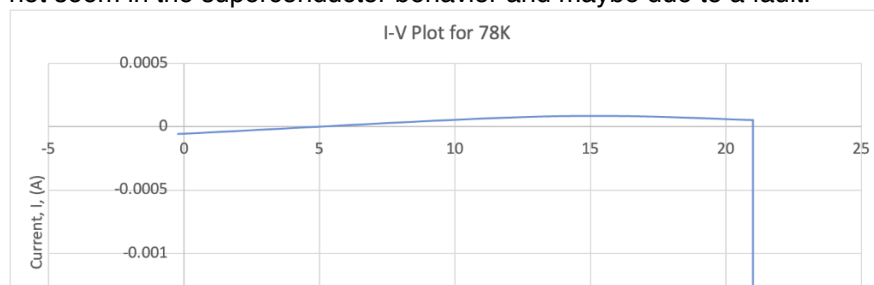
The resistance at 95K is $9 \times 10^{-6} \Omega$. Data extremely unstable. At this point tip is completely submerged and evaporation is causing fluctuations in the data.

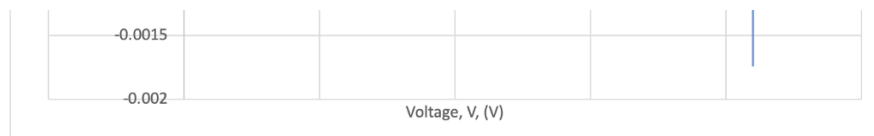


At 85K, Resistance= $5.5 \times 10^{-6} \Omega$. Unstable data continues.



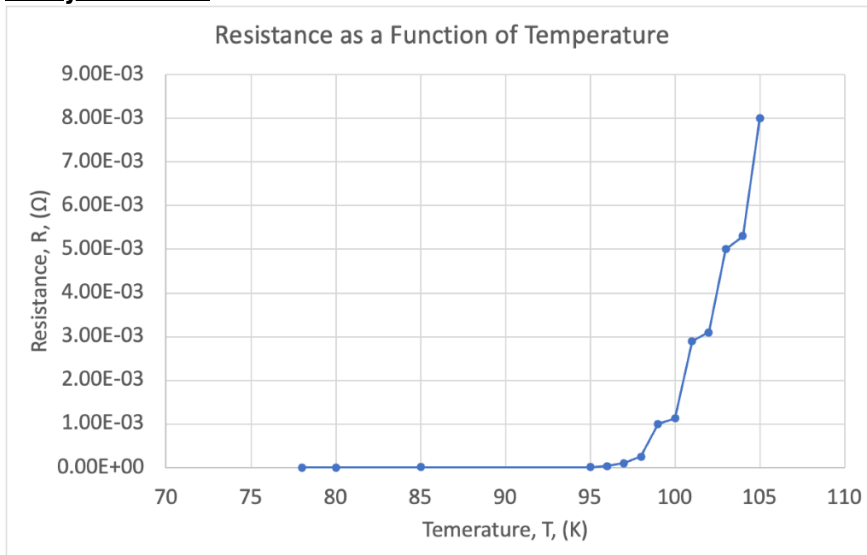
Resistance for 80K is almost 0Ω . The spike in current at around 20V does not seem in the superconductor behavior and maybe due to a fault.



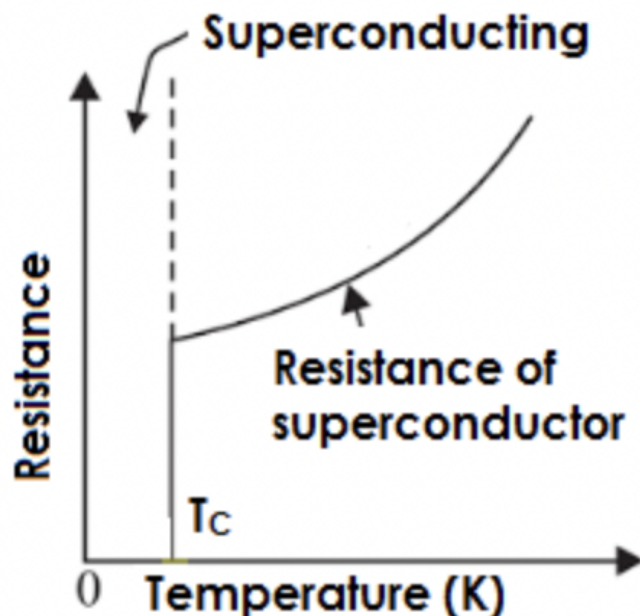


Resistance at 78K is almost 0 and current has increased, superconductivity has been achieved in practice.

Analysis task 2:



The curve for this graph does demonstrate the main idea of T against R in that resistance does decrease as temperature decreases. They are proportional. However, the curve does not find that in the relevant literature. This is partly due to the lack of points which makes the curve jagged. But more than this, a super conductor curve should have a sudden drop of at a specific temperature to 0, instead this graph gradually goes to 0. the only part that could be seen as this drop off is around 98K.



Above is the R-T graph for a typical superconductor.

This difference in behavior/error may be due to the evaporation of liquid nitrogen around the tip of the probe. This was observed in the lab. This would have an effect on the tip temperature which also explains the instability at the lower temp I-V graphs.

A fix for this would be to let the liquid nitrogen stabilize for longer once the probe is submerged. We only stabilized for 5-10 minutes due to lack of

time. As well as this a larger container of liquid nitrogen may reduce the temperature change in the container as a whole.

As well as this errors can arise from the instruments used to measure electrical resistance, magnetic field, or other relevant quantities.

Instrumental noise, limitations in measurement sensitivity, calibration errors, or inappropriate measurement techniques can impact the accuracy of the obtained data. Making sure all equipment is checked and calibrated is vital to the accuracy of the data

Compton Scattering – 16/12/2022

Monday, May 08, 2023 10:11 PM

Task 1:

MCA Calibration.

To calibrate:

1. Make sure the path to the detector is not blocked by the acrylic glass sample.
2. Insert the tube with the 1 mm-aperture into the exit tube of the X-ray tube.
3. Close door and lock the X-ray machine, then set the detector angle to 0°.
4. Within the instrument's menu, go to X-ray settings and set an acceleration voltage of 25 kV and a current of 0.02 mA.
5. Switch on the radiation by pressing the radiation button.
6. In the "Measure" program select the "Spectra Recording" to measure a spectra.
7. Locate the current counts variable. If the number is above 500 counts/s adjust the detector position until the value is at or below 500 counts/s. Then cancel the measurement.
8. In the "Measure" program select the Settings and Calibration option and click "Calibrate".
9. Select: "2-point Calibration", Unit keV, Gain= 1, and an offset that eliminates the low energy peak.
The detector will now measure Impulses (which is the number of x-rays it has detected) vs. Channel# (which are the uncalibrated units of energy).
10. Accumulate counts until you can identify the K α 1 and K β 1 peaks
What other spectral features can you see?
11. Move the two colored calibration markers to be congruent with the centers of the two characteristic X-ray peaks. Enter the corresponding energies for the peaks.
12. Click "Apply"
13. "close"

Task 2:

Spectrum of the x-ray source.

measure a spectrum of the Molybdenum x-rays. In order to collect high quality data, the acceleration voltage and current need to be increased in the X-ray settings menu. You should also make sure that the detector is placed around mid-way in its guiding rack.

1. Change X-ray generator settings to 30 kV and 0.3 mA.
2. put in the 5 mm aperture into the X-ray output slot.
3. Close the door and lock the instrument.
4. Move the detector to 10° and switch on the radiation.
5. The acquisition settings should be Unit keV, Gain= 1, Interval width [channels]= 1, Offset as you used in calibration.
6. Observe the number of counts and adjust the detector position until the count rate is $\leq 500 \text{ counts s}^{-1}$.

7. Reset the measurement.
8. Accumulate counts until you have a clear spectrum.
9. Save you spectrum.

The spectrum will be good if it is clean and smooth, not scattered, and resembles the two peak display we are familiar with.

Energy range, resolution, sensitivity, sampling rate, calibration and accuracy will all be a part of criteria throughout the experiment.

Task 3:

Inelastic scattering of X-rays.

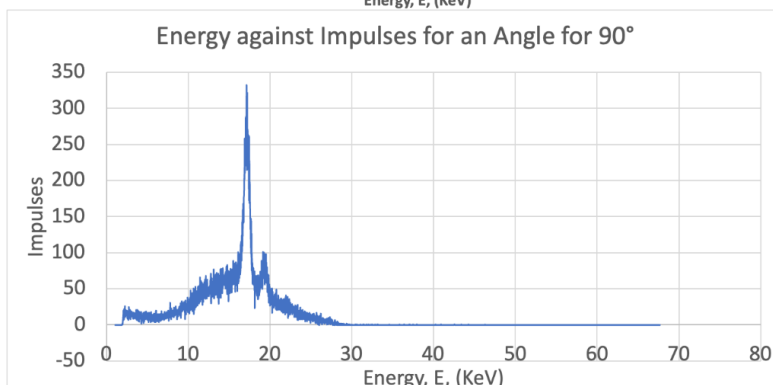
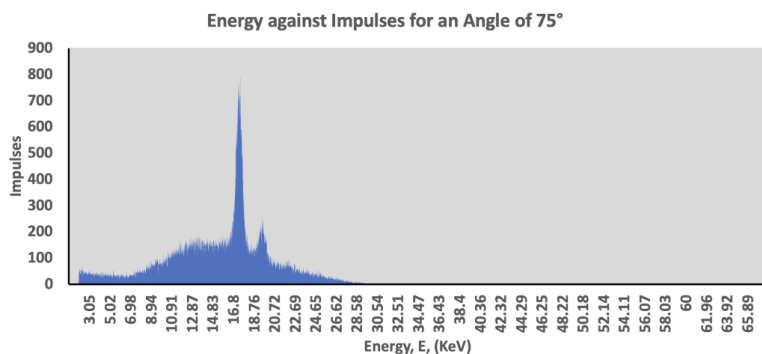
To observe the Compton scattering effect, we must introduce a source of weakly bound electrons for the X-ray beam to interact with. This will be achieved in this experiment with the acrylic glass.

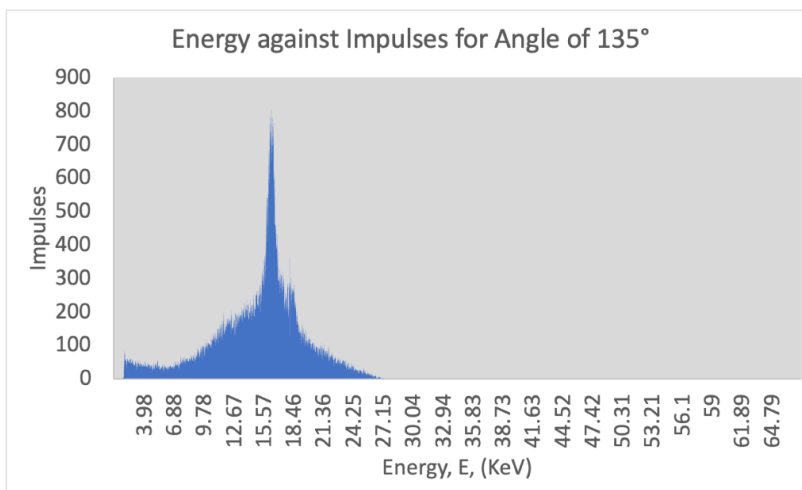
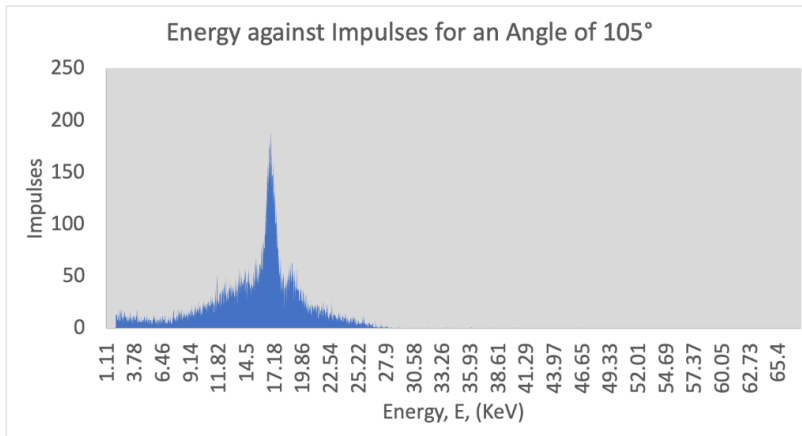
1. Place the acrylic glass in the sample holder at the center of the goniometer.
2. Close the door and lock the instrument.
3. Turn the X-ray source on and place the sample angle to 10° and the detector angle to 30° .
4. Collect data until you have a clear spectrum where you can identify the main peaks. Save each measurement including the position of the detector in degrees.
5. Vary the position of the goniometer arm between 30° and 150° in 15° steps recording spectra as you go.
6. Once the last spectrum is acquired, turn the radiation source off, unlock the instrument.

The count rate may change depending on the source intensity, due to the angle. As well as instrument stability, sample composition, absorption and scattering effects, background radiation.

105degs has lowest count rate.

Analysis task 1:





The origin of the additional peaks are background radiation and stability of the machine as well as sample composition being ununiform.

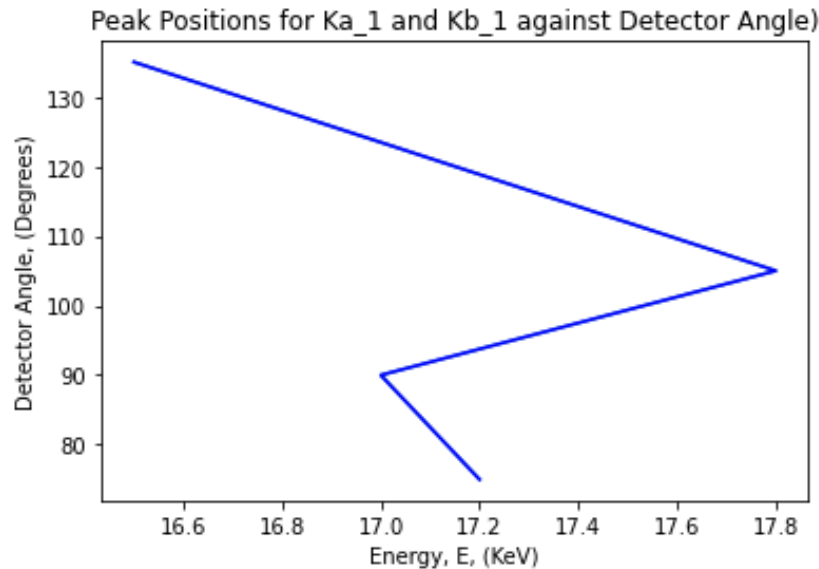
The distribution of energies for a single emission peak can arise from a phenomenon known as spectral broadening or line broadening. Spectral broadening occurs when the emitted photons or particles possess a range of energies rather than being emitted at a single, discrete energy level.

A Gaussian distribution is often used to fit peaks because it has several desirable properties such as symmetry and central limit theorem.

To improve the accuracy of the measurements conducting the experiment in a sealed container to the outside world to protect from back ground radiation. As well as this, making sure the sample is well packed and smoothly finished at the top as well and getting

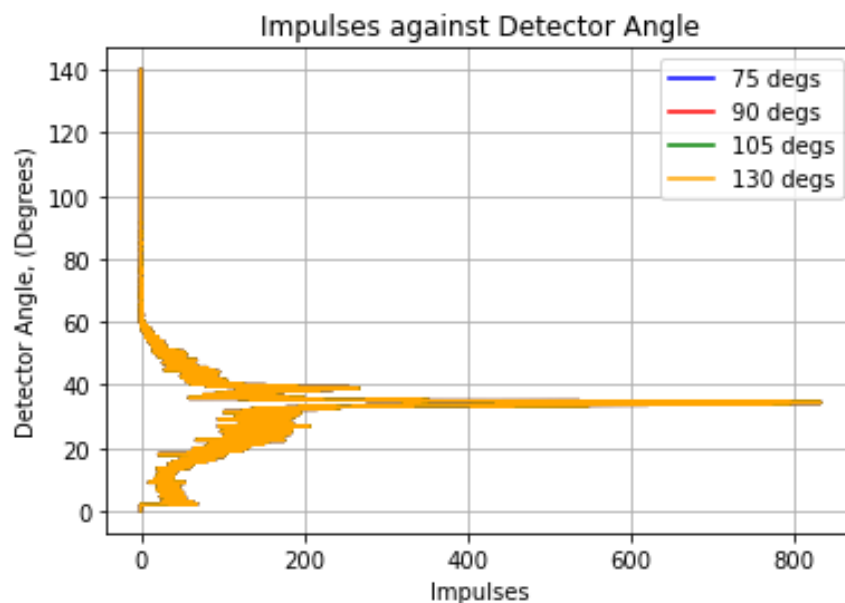
rid of any impurities would also help.

Analysis task 2:



As you can see both Ka₁ and Kb₁ peaks are rising and falling around the same energy values, the maximum peak is greatest at an angle of 105°. The energy rises and falls over the range of the detector angle. Which is expected.

Analysis task 3:



There is a similar pattern to the energy against detector angle. Where there is a sharp increase. However, at a different angle, this time it is around the 30 degrees mark. This is because there are more X-rays refracting and hitting the detector at this angle.

The overall error for this experiment is moderate. The samples used were not verified and checked for imperfections/impurities etc. - Next time, having pre-checked samples handled with gloves in an controlled environment would minimize this error.

The background radiation was not controlled/blocked. And was not accounted for or subtracted during the data analysis. - Next time having the experiment completed in a protected environment from environmental radiation would reduce this error and also implementing edited data analysis to account for this extra signals would also help reduce noise.

The Detector used to measure the scattered radiation may have inherent limitations such as noise, dead time, or non-uniform response. These factors can introduce uncertainties in the measured data. - next time, understanding and noting the limitations of the detector before carrying out the experiment and keeping the proposed experiment within the boundaries of what the machine can do would allow the data to be more accurate and conclusive.

In some cases, multiple scattering events can occur, where a single photon undergoes multiple scattering interactions before being detected. These multiple scattering events can complicate the data analysis and introduce uncertainties. - it is hard to account for this in the data, especially with the equipment used. But understanding this this could take place and may lead to some of the uncertainty in the data allows a more holistic understanding of the system in the experiment.

Hall Effect 7/2/2023

Friday, March 3, 2023 4:01 PM

Set up:

1. Power supply:

The power supply provides both AC and DC power.

AC: The AC supplies are at a fixed voltage determined by the placement of the jumper on a dial. Each combination is indicated on the face of the dial. The AC supply is for the Hall Effect module; make sure that you are supplying enough voltage for it to function correctly.

DC: The DC supplies power using current/voltage limiting. This means that the power output will be limited by the value of V or I that results in a smaller power output. You can tell which value is limiting the power output by the indicator light which lights up if the values are current limited.

2. Electromagnetic coil and core:

There are two electromagnetic coils that are designed to be used with a magnetic core where each coil defines a single pole of the magnetic field. The three parts of the core are placed so that the magnetic field is applied perpendicular to the sample card when mounted in the Hall Effect module. The DC output from the power supply drives both coils simultaneously, so you must make sure that their polarity is correct.

3. Hall effect module:

The Hall Effect module is mounted on the retort stand such that it sits vertically and the Ge sample in the Sample card is situated between the poles of the electromagnet. The Sample card is loaded into the Hall Effect module via a guide groove. Be careful when mounting this piece as it is possible to damage the card. The Hall Effect module takes a 12 V AC power supply to drive current through the hall sample.

4. Sample card:

The Sample card is loaded into the Hall Effect module with the pins uppermost. The Hall module and Sample card are loaded into the Electromagnet poles from the top down. Please note the number on the card in your lab book.

5. Hall probe and Teslameter:

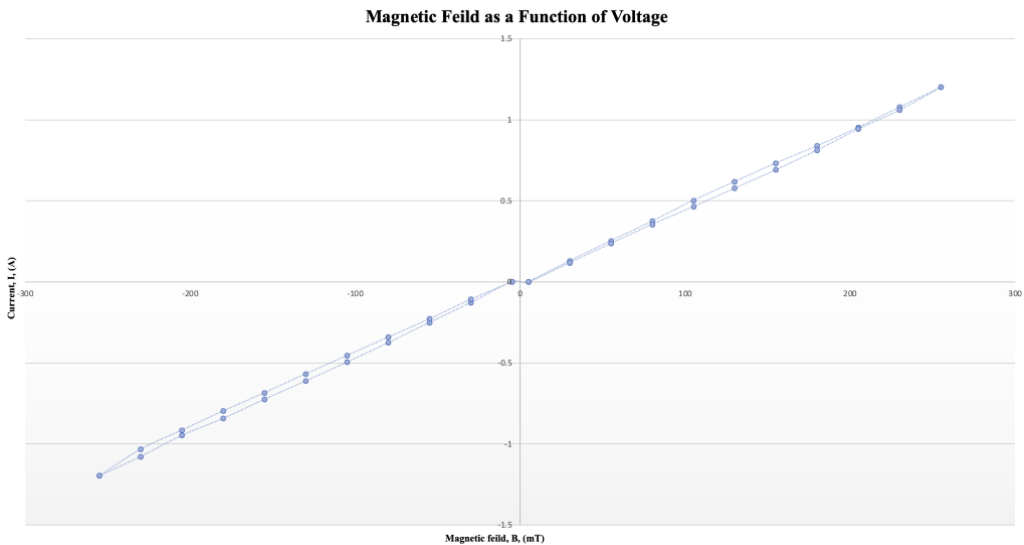
The hall probe is a tool for measuring the magnetic field. It is inserted into the hole at the top of the Hall Effect module and the flat end is located next to the Ge sample on the Sample card. Be careful that the flat end does not damage the delicate connection to the Ge sample. The hall probe D-connector plugs into the telemeter which will read out the field strength accurate in the millitesla range. Make sure that you understand what direction the field is being measured by the probe.

6. Multimeter:

You will use the multimeter to measure all values in this experiment except for IP and B. Make sure you know the polarity of the probes (i.e. Cathode-Hi-red, Anode-Lo-Black) and use the correct wiring to make the measurement you want.

Task 1:

Deducing the characteristics of the electromagnet. Measure B versus IC for B at 25 mT intervals from 0 mT → 250 mT → 0 mT → -250 mT → 0 mT (reverse cables when IC=0)



B(mT)(±0.5mT)	I(A)(±0.5mA)
5	0
30	0.129
55	0.253
80	0.374
105	0.502
130	0.618
155	0.732
180	0.839
205	0.953
230	1.077
255	1.2
255	1.2
230	1.059
205	0.942
180	0.812
155	0.691
130	0.577
105	0.463
80	0.354
55	0.237
30	0.117
5	0
-5	0
-30	-0.129
-55	-0.251
-80	-0.376
-105	-0.496
-130	-0.612
-155	-0.725
-180	-0.843
-205	-0.948
-230	-1.078

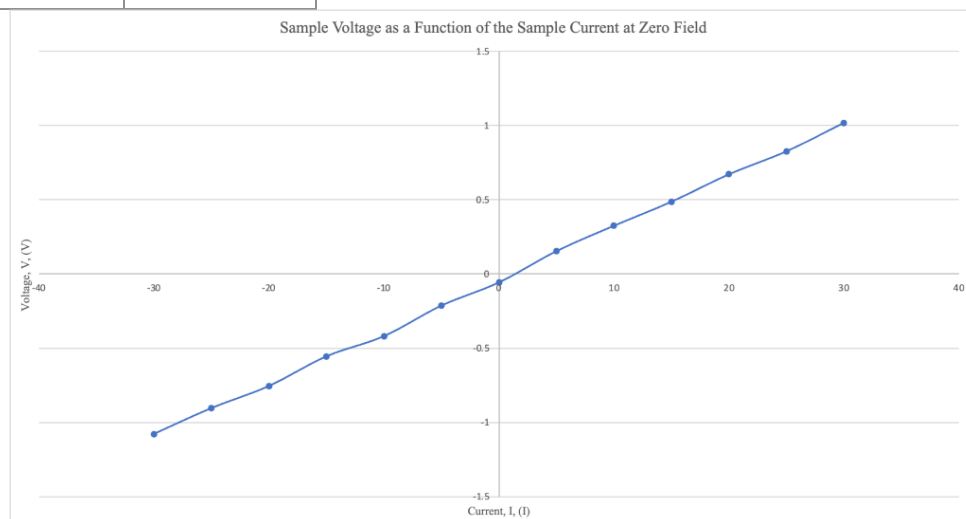
-255	-1.197
-255	-1.197
-230	-1.033
-205	-0.915
-180	-0.797
-155	-0.685
-130	-0.568
-105	-0.455
-80	-0.342
-55	-0.229
-30	-0.109
-5	0

Task 2:

Deduce conductivity of Germanium. Connect the multimeter to the sockets of the sample voltage on the front-side of the module.

Determine the sample voltage as a function of the sample current from -30 mA up to 30 mA in steps of 5 mA.

I(A)(±0.5mA)	V(V)(±0.0005V)
-30	-1.079
-25	-0.905
-20	-0.756
-15	-0.556
-10	-0.42
-5	-0.214
0	-0.058
5	0.153
10	0.324
15	0.486
20	0.671
25	0.826
30	1.018



the relationship between sample current and voltage is directly proportional. Finding the gradient of this curve along with the initial measurements of the sample allowed the use of equation to be used to determine the conductivity of Germanium.

$$\sigma = \left[\left(\frac{l}{A} \right) \left(\frac{V}{I} \right) \right]^{-1}$$

$$(d = 10^{-3}m, l = 0.02m, w = 0.01m)$$

Conductivity of germanium was found to be 0.286 Sm^{-1}

Task 3:

Determine the relationship between the sample current and Hall voltage.

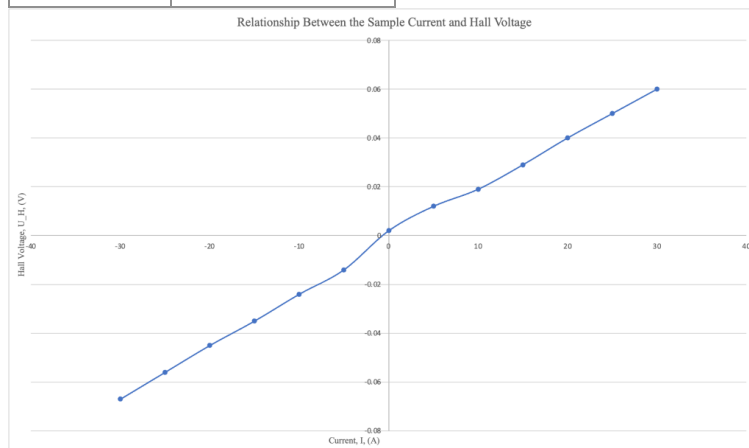
Determine

the proportionality factor. Using this result estimate the charge carrier concentration in the sample.

Task

Set the magnetic field to a value of 250 mT by changing the voltage and current on the DC power supply. Connect the multimeter to the sockets of the hall voltage (UH) on the front side of the module. Set the display on the module into the "current-mode". Determine the hall voltage as a function of the sample current from -30 mA up to 30 mA in steps of 5 mA.

I(A)(±0.5mA)	U _H (V)(±0.0005V)
-30	-0.067
-25	-0.056
-20	-0.045
-15	-0.035
-10	-0.024
-5	-0.014
0	0.002
5	0.012
10	0.019
15	0.029
20	0.04
25	0.05
30	0.06



The determination of the relationship between Hall voltage and current allowed proportionality factor to be calculated. The relationship is directly proportional and the factor was calculated to be 0.002. The equation was used to determine charge carrier concentration in conjunction with the proportionality factor which corresponded with the I/U_H out the front of equation.

$$n = \frac{I}{U_H} \left(\frac{B}{ed} \right)$$

B=250mT.

n was calculated to be $n=3.211 \times 10^{18} \text{ m}^{-3}$

Task 4:

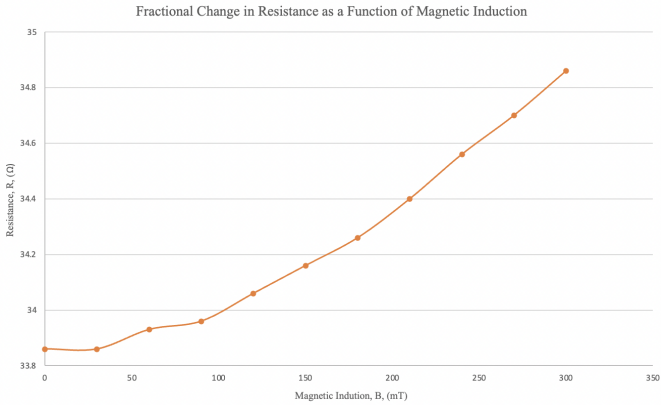
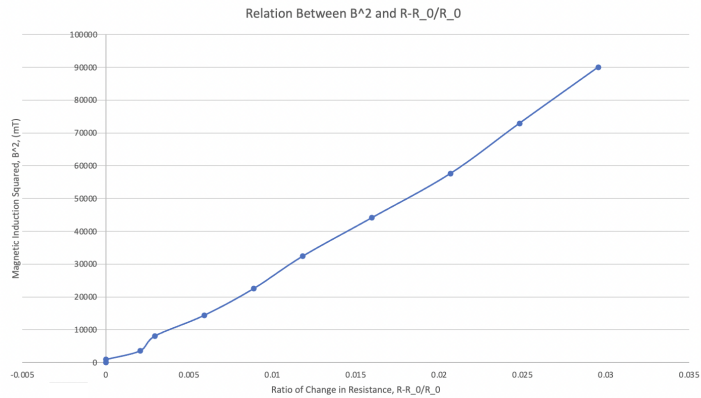
Verify the relation in Equation:

$$\frac{R-R_0}{R_0} \propto B^2$$

Set the sample current (IP) to 30 mA. Connect the multimeter to the sockets of the sample voltage on the front side of the module. Determine the sample voltage as a function of the positive magnetic induction B up to 300 mT.

B(mT)(±0.5mT)	V(V)(±0.0005V)
0	1.016
30	1.016
60	1.018

90	1.019
120	1.022
150	1.025
180	1.028
210	1.032
240	1.037
270	1.041
300	1.046



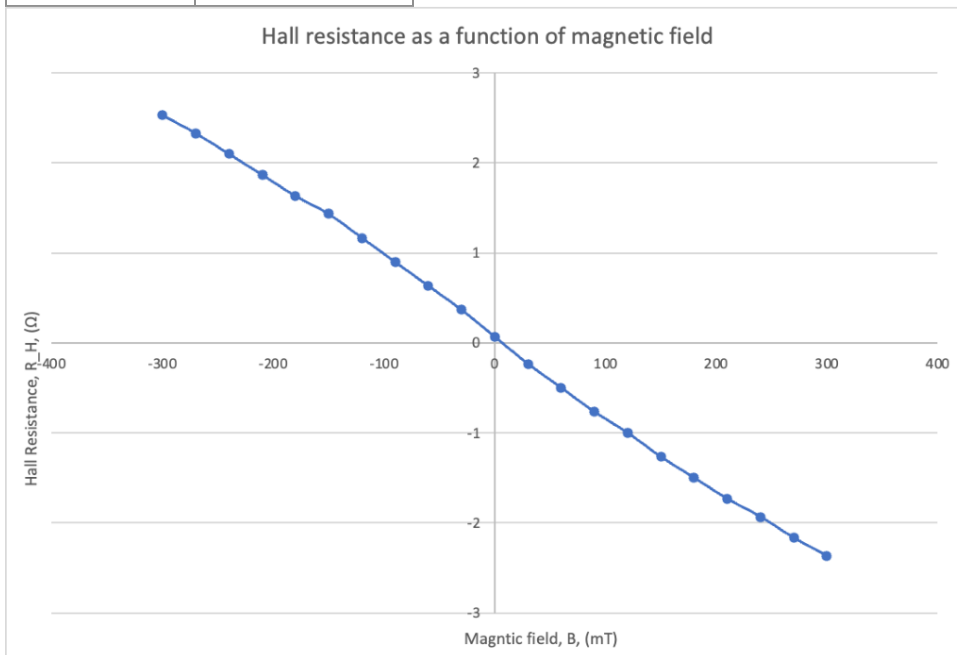
Relation holds, graphs are a good fitting of each other.

Task 5:

Plot the Hall resistance as a function of magnetic field.
Is this sample n- or p-type?
Calculate the carrier concentration in the sample. Compare this value with the carrier concentration in Task 3. Which is more accurate and why?
Calculate the sample mobility.

B(mT)(±0.5mT)	Hall resistance(Ω)
-300	2.53
-270	2.33
-240	2.1
-210	1.867
-180	1.633
-150	1.433
-120	1.167
-90	0.9
-60	0.633
-30	0.367
0	0.0667
30	-0.233
60	-0.5
90	-0.7667
120	-1
150	-1.2667

180	-1.5
210	-1.733
240	-1.9332
270	-2.167
300	-2.3667



Semiconductor = n-type, negative slope, (discussed with Alex Johnson)

$n = 4.588 \times 10^{17} \text{ m}^{-3}$, It is expected that the old n is more accurate, due to the constant current which did not change the remnant magnetic field.

Carrier mobility was calculated to be $3665.962 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$

Potential sources of inaccuracy:

Magnetic field calibration: Accurate calibration of the magnetic field strength is crucial for reliable measurements. Errors in the magnetic field calibration can lead to inaccuracies in determining the Hall voltage and other related parameters. Calibration should be performed using reliable and well-calibrated magnetic field sensors. - Next time calibration and equipment testing before the experiment will help reduce error.

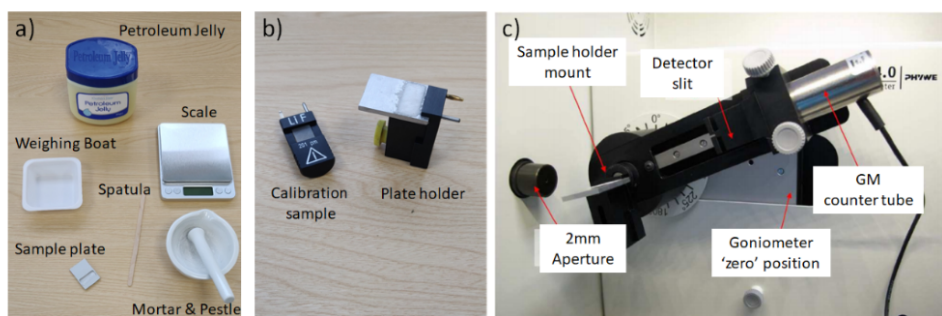
Electrical noise: Electrical noise can interfere with the Hall voltage measurement. Sources of noise can include power supply noise, fluctuations in the measuring circuit, or external electromagnetic interference. -Next time proper grounding, shielding, and filtering techniques should be employed to minimize noise and improve the signal-to-noise ratio.

XRD – 31/1/2023

Tuesday, January 31, 2023 10:23 AM

Setup:

- X-Ray diffractor unit – allows X-rays to be generated for experiment.
- Goniometer – directs beam and changes beam angle on the sample.
- Sample holder + sample holder mount – hold the crystal/Vaseline sample in the unit, sample holder filled flush with top surface.
- Aperture – 2mm and 4cm away from sample detector according to lab script.
- Sample and detector are 10cm apart according to lab script.
- All distances are to be checked with ruler to allow accuracy.



Task 1:

Measure out 2g of NaCl and grind with mortar and pestle then back into weighing boat and mix with 0.3g of Vaseline to get a dry consistency. Once done fill compartment with substance making sure it is level with top of compartment. We found using side of the wooden mixing stick to get it level.

Task 2:

The goniometer needs to be calibrated to ensure that the data is in line with expected values to ensure accurate experiment data. This was done using a set LiF calibration sample. The calibration program is then run on the measure program on the computer. Takes about 10 minutes.

Calibrating the X-ray detector is necessary to ensure accurate and precise measurements of X-ray intensity. During calibration, various parameters such as gain, offset, linearity, and energy response of the detector are measured and adjusted to achieve reliable and accurate readings.

Task 3:

Scanning the sample that has been placed into the holder is next. Once door is locked the settings should be changed.

- Mode – 1:2 coupling mode
- Crystal start angle – 5 deg
- Crystal stop angle – 20 deg
- Crystal increment – 0.1 deg
- Integration time – 20s

Once set recording takes place and takes 50 minutes.

Data is saved to device. (aperture distances saved for later calculation)

Best angular resolution:

$$\text{Angle} = 2 \cdot \text{wavelength} / dA \cdot L$$

(wavelength=0.07nm, dA=2mm, L=10-4=6cm, all values are specified in script)

$$\text{Angular resolution} = 0.000002447^\circ$$

Molybdenum has X-ray source has multiple emission peaks. This has an effect on the accuracy of the measurements. Firstly the presence of multiple emission peaks can result in peak overlap in the diffraction pattern, which can make it difficult to accurately measure the peak positions and intensities. Secondly, the relative intensities of the different emission lines can vary, which can affect the overall intensity of the diffraction peaks. The $K\alpha_1$ and $K\alpha_2$ lines, for example, have similar energies but different relative intensities, which can cause $K\alpha$ peak in the diffraction pattern to be asymmetric.

Bragg's law:

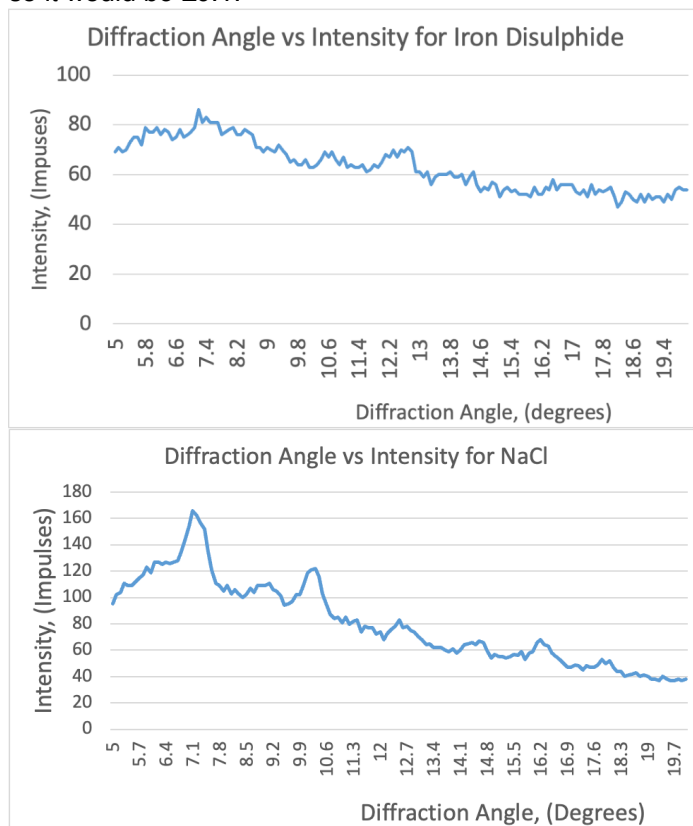
$$2 \cdot d \cdot \sin(\text{angle}) = n \cdot \text{wavelength}$$

Diffraction peaks for $K\alpha$ and $K\beta$ are about 1.6° apart

Task 4:

Making a plot of intensity vs angle in measure was next.

The uncertainty in these measurements are ± 1 of the least significant figure so it would be ± 0.1 .



Each peak is an individual Debye-Scherrer ring that can be assigned to corresponding lattice planes. As you can see the peaks are not very clear. This may be due to equipment error in the form of sensitivity or a general fault with connections etc. Or most likely human error in the form of packing the samples. It is a very involved process with lots of different potential errors such as smoothing the surface, packing density, internal distribution of the sample, etc.

For Iron disulphide: glancing angles are at approximately 7.4° and 12.8° .

$\frac{\sin^2(a_n)}{\sin^2(a_{min})} = 0.06635926175$, it takes x3 to make into an integer therefore it is a face centered cubic.

There is some error in this however, in the order of 0.01. This is less than the expected error of ± 0.1 . therefore it is expected that this error has arisen.

For NaCl: glancing angles are at approximately 4.1° , 10° , 12.7° , 14.6° and 16.2° .

$\frac{\sin(a_n)^2}{\sin^2(a_{min})} = 1.19576018$, it takes x3 to make into an integer, with the same error as the iron disulphide, therefore it is a face centered cubic.

Task 5:

Determining the plane spacing, d , and the lattice constant, a , for each peak.

$d = n(\text{wavelength}) / 2\sin(\text{angle})$
 $a = ((\text{wavelength})^2 / 2\sin(\text{angle})^2)(h^2 + k^2 + l^2)$
 Assume wavelength to be 1.515nm

For Iron disulphide:

$d_1 = 5.88 \times 10^{-9}\text{m}$, $d_2 = 6.84 \times 10^{-9}\text{m}$
 $a_1 = 2.1 \times 10^{-16}\text{m}$, $a_2 = 7.01 \times 10^{-17}\text{m}$

The estimate for a from the results would be closer to a_1 as that is closer to the order of tabulated data on crystal structures. The uncertainty could be reduced if calculations were not rounded and peaks were not picked out by eye, instead a piece of code programmed to do this job would be more accurate given that the samples were correctly made.

For NaCl:

$d_1 = 1.059 \times 10^{-8}\text{m}$, $d_2 = 8.7245 \times 10^{-9}\text{m}$, $d_3 = 1.034 \times 10^{-8}\text{m}$,
 $d_4 = 1.202 \times 10^{-8}\text{m}$, $d_5 = 1.358 \times 10^{-8}\text{m}$
 $a_1 = 6.73 \times 10^{-16}\text{m}$, $a_2 = 1.14 \times 10^{-16}\text{m}$, $a_3 = 7.123 \times 10^{-17}\text{m}$,
 $a_4 = 5.42 \times 10^{-17}\text{m}$, $a_5 = 4.423 \times 10^{-17}\text{m}$

The estimate for a would be closer to a_2 for the same reason as the iron disulphide. It is closer to the tabulated data on crystal structures in the range of an angstrom.

Potential sources of inaccuracy:

Sample preparation: Improper sample preparation can introduce errors. Factors such as the size, shape, orientation, and surface condition of the sample can affect the diffraction pattern. Irregularities in sample grinding, uneven thickness, and residual stress can lead to incorrect peak positions, intensities, or broadening. - next time making multiple samples and conducting repeats of the experiment will help reduce the inaccuracy. As well as having collaboration with other scientists to check samples and look for imperfections.

The background radiation was not controlled/blocked. And was not accounted for or subtracted during the data analysis. - Next time having the experiment completed in a protected environment from environmental radiation would reduce this error and also implementing edited data analysis to account for this extra signals would also help reduce noise.

The Detector used to measure the scattered radiation may have inherent limitations such as noise, dead time, or non-uniform response. These factors can introduce uncertainties in the measured data. - next time, understanding and noting the limitations of the detector before carrying out the experiment and keeping the proposed experiment within the boundaries of what the machine can do would allow the data to be more accurate and conclusive.

The Ising Model – 14/2/2023

Friday, May 12, 2023 2:54 PM

Simulation set up code:

```
import numpy as np
import matplotlib.pyplot as plt

def initialize_spins(N):
    return np.random.choice([-1, 1], size=N)

def calculate_energy(spins, J):
    N = len(spins)
    energy = 0
    for i in range(N):
        energy += -J * spins[i] * spins[(i+1) % N]
    return energy

def metropolis_update(spins, J, temperature):
    N=len(spins)
    i = np.random.randint(N)
    dE = 2*J*spins[i]*(spins[(i-1)%N]+spins[(i+1)%N])
    if dE <= 0 or np.random.rand() < np.exp(-dE/temperature):
        spins[i]*=-1

    return spins

def run_ising_model(N, J, temperature, num_steps):
    spins=initialize_spins(N)
    spins_history = [spins.copy()]
    energy_history = [calculate_energy(spins, J)]
    for step in range(num_steps):
        spins = metropolis_update(spins, J, temperature)
        spins_history.append(spins.copy())
        energy_history.append(calculate_energy(spins, J))
    return spins_history, energy_history

N = 100
J = 1
temperature = 2
num_steps = 1000

spins_history, energy_history = run_ising_model(N, J, temperature,
num_steps)

plt.figure(figsize=(10,6))
plt.subplot(211)
plt.imshow(np.array(spins_history).T, cmap='binary', aspect='auto')
plt.xlabel("Time step")
plt.ylabel("Spin index")
plt.title("Spin Configuration over Time")
plt.colorbar()

energy_array = np.array(energy_history)
```

```

window_size = 10
smoothed_energy = np.convolve(energy_array,
np.ones(window_size)/window_size, mode='valid')
energy_derivative = np.gradient(smoothed_energy)
max_derivative_index = np.argmax(np.abs(energy_derivative))
critical_temperature = temperature - (num_steps - window_size) /
num_steps*temperature
print(critical_temperature)

```

```

def calculate_heat_capacity(energy, temperature):
    mean_energy = np.mean(energy)
    mean_energy_squared = np.mean(energy**2)
    beta = 1 /temperature
    heat_capacity = (beta**2) * (mean_energy_squared -
(mean_energy**2))
    return heat_capacity

```

```

energy_array = np.array(energy_history)
temperature_array = np.array(temperature)

```

```

heat_capacity_array = calculate_heat_capacity(energy_array,
temperature_array)

```

```

plt.plot(temperature_array, heat_capacity_array)
plt.xlabel('Temperature, T, (K)')
plt.ylabel('Heat Capacity, C_v, (JK^-1)')
plt.title('Heat Capacity vs Temperature')
plt.show()

```

Temperature dependance:

After running the simulation with the inputted values the critical temperature was calculated to be: 0.02K

However, the Ising model simulation and the determination of the critical temperature are idealized and simplified representations. In real-world systems, the critical temperature may vary and depend on various factors such as the material properties, interactions, and other external conditions. The Ising model serves as a basic model to understand certain aspects of phase transitions and critical phenomena. Therefore it is not totally accurate to nature but does allow understanding.

Magnetic field dependance:

In the presence of a magnetic field, the behavior of the system in the Ising model can change significantly. The magnetic field introduces an additional interaction term in the energy equation, affecting the alignment of the spins and altering the equilibrium properties of the system.

In the Ising model, each spin can have two possible states: up (+1) or down (-1). The spins interact with their nearest neighbors, and the energy of the system is determined by the alignment of neighboring spins. When a magnetic field is applied, it introduces an external influence on each spin.

Overall, the magnetic field modifies the energy landscape of the Ising model, leading to changes in the spin alignment, magnetization, and system behavior. The specific effects depend on the field strength, interaction parameters, and temperature, and can be further explored through simulations or analytical calculations.

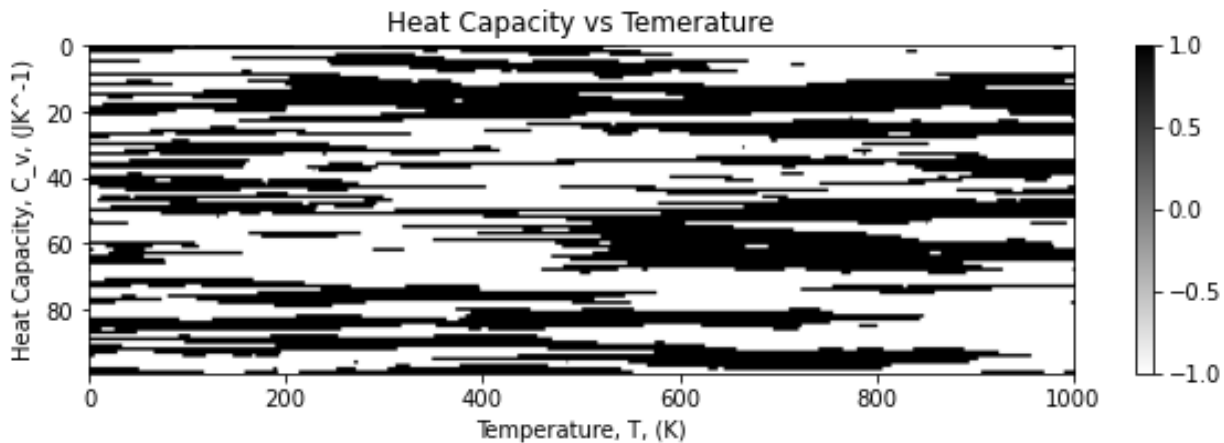
"does the system demonstrate a phase change at $T = 0$, at $T = 5$?"

At $T = 0$: When the temperature is at absolute zero, the system is in its lowest energy state. In this case, the spins tend to align to minimize the

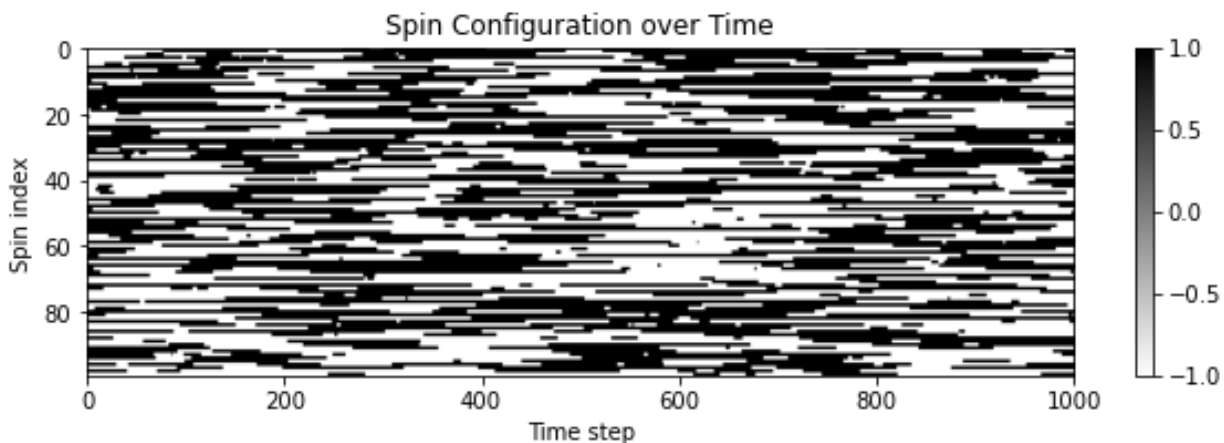
energy and maximize the number of parallel neighboring spins. The system exhibits long-range order, where all spins are aligned either up or down, depending on the initial conditions. This phase is often referred to as a ferromagnetic phase.

At $T = 5$: the system is at a finite temperature above absolute zero. The behavior of the system depends on the interaction strength (J) and the applied magnetic field (B), in addition to the temperature. The phase transition behavior at this temperature may vary depending on these factors.

Heat capacity against temperature:



Spin configuration for the system:



Sources of inaccuracy/uncertainty:

Finite simulation time: The number of simulation steps may not be sufficient for the system to reach equilibrium or to fully explore the phase space. Insufficient simulation time can result in inaccurate estimation of properties, especially near the critical temperature where the system exhibits slow dynamics – allowing the simulation sufficient time would help reduce the uncertainty.

Numerical precision: Numbers in a computer has finite precision, which can introduce rounding errors or truncation errors. These errors can accumulate over time and affect the accuracy of calculations, especially for long simulation runs or when dealing with small energy differences – accounting for this within the data analysis would help reduce this.

Alternating Current Circuits – 3/7/2023

Wednesday, May 10, 2023 8:33 PM

Setup:

Equipment:

Wave form generator: A waveform generator capable of producing sine, square and triangular waveforms is provided. The instrument can introduce an 'offset' to the output wave from -20 V to 20 V , while the amplitude is limited to a maximum of 10 V .

Oscilloscope: A standard oscilloscope is provided. Note that sensitive probes are provided that look like a 'crocodile clip' as well as a small 'hook'. The crocodile part of the probe should be connected to the ground wire of the circuit, while the 'hook' is used to probe the component.

Breadboard: A solder-less breadboard.

1. Use a BNC cable to connect the 'TTL' port on the function generator to the 'Ext Trig' port in the oscilloscope, and in the oscilloscope 'Trigger Menu', set the source to 'EXT'.¹²³
2. Connect the T-connector to the wave-generator output port and plug in the BNC adapter on one side and connect the yellow RF power cable to this port.
3. Use a BNC cable to connect the free port on the T-connector to 'Channel 1' on the oscilloscope. Hit the yellow '1' button of Channel 1 on the oscilloscope and set the attenuation to 'x1': Probe $\rightarrow$ attenuation $\rightarrow$ x1. Do the same for Channel 2 by pressing the blue '2' button. Select Channel1 again.
4. On the wave generator, select 'sine wave' and turn on the output
5. Press the blue 'Shift' button and immediately press the 'TTL' button to turn on the TTL output.
6. Measure the resulting waveform on the channel 1 of the oscilloscope by pushing the 'Measure' button and set the measurements to Channel 1 'mean' and Channel 1 'Pk to Pk'.
7. Tweak the amplitude and offset controls to set a waveform with 1 V amplitude and 0 V offset at 1 kHz .

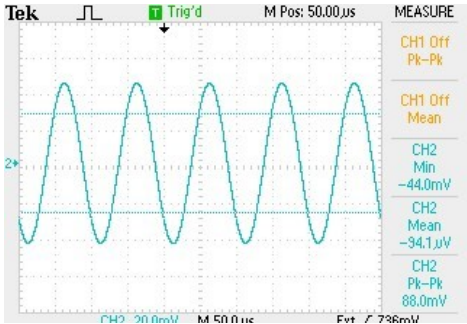
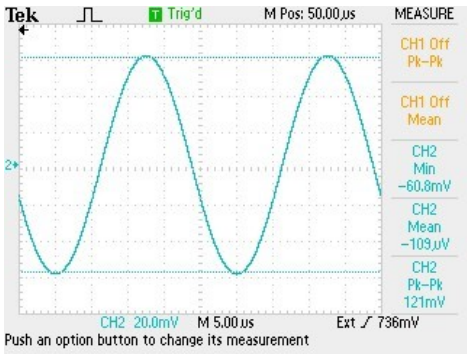
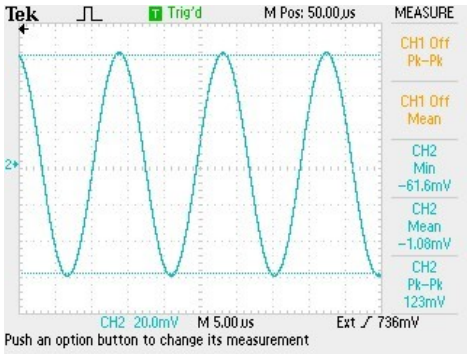
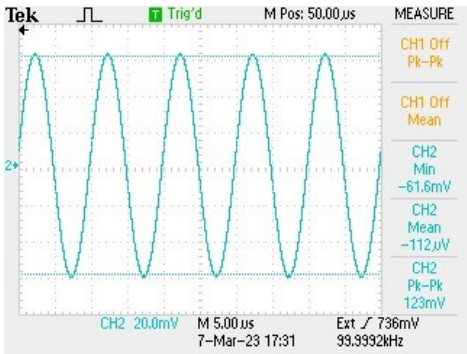
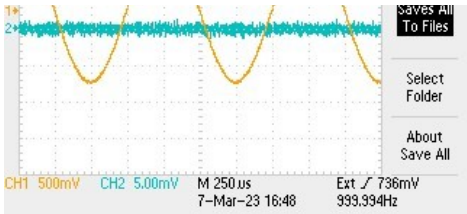
Task 1:

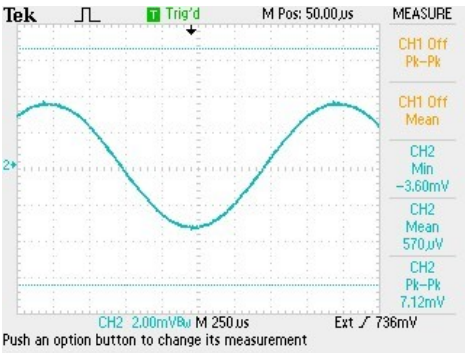
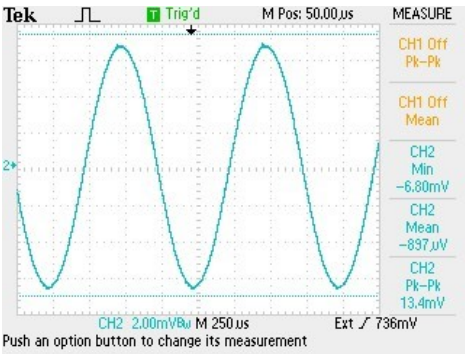
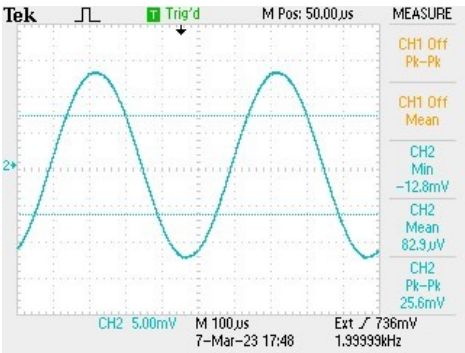
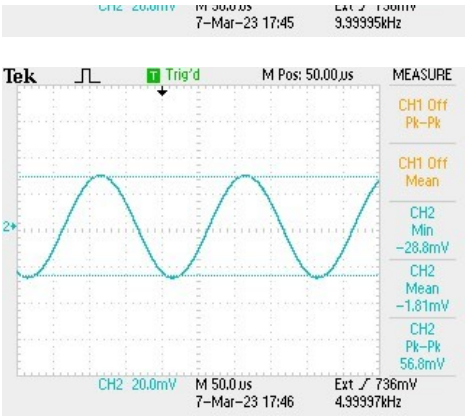
Measure the waveforms of the voltage across the resistor against frequency from a range of 100 kHz down to 1 Hz in suitable steps and save each on the USB stick.

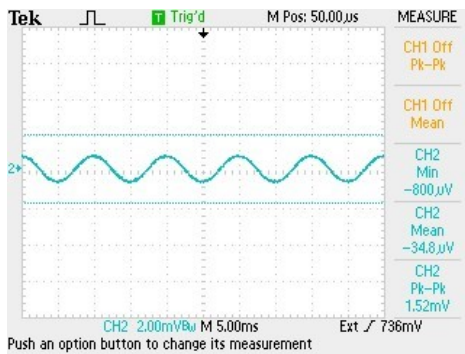
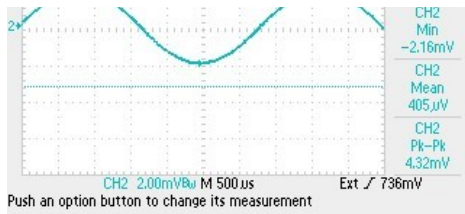
RC Circuit:

Frequency	PEAK to PEAK	$\Delta V_{A,B}$	$\Delta V_{C,D}$
100 kHz	123 mV	125 mV	122 mV
40 kHz	123 mV	125 mV	120 mV
40 kHz	122 mV	122 mV	118 mV
20 kHz	111 mV	113 mV	109 mV
10 kHz	86 mV	89.6 mV	86.4 mV
5 kHz	56.8 mV	57.6 mV	54.4 mV
3 kHz	36.8 mV	37 mV	36 mV
2 kHz	25.6 mV	26 mV	24.8 mV
1 kHz	13.4 mV	13.7 mV	13 mV
500 Hz (0.5 kHz)	7.12 mV	7.12 mV	6.56 mV
300 Hz (0.3 kHz)	4.4 mV	4.32 mV	3.76 mV
100 Hz (0.1 kHz)	1.52 mV	1.6 mV	1.2 mV









Range of waveforms from 100KHz (top), 70KHz, 40KHz, 20KHz, 10KHz, 5KHz, 3KHz, 2KHz, 1KHz, 500Hz, 300Hz, 100Hz, 50Hz

Analysis task 1:

Using Python to extract the capacitance of the RC circuit.

Code:

```
-----
import numpy as np
from scipy.optimize import curve_fit

frequencies = np.array([100, 200, 300, 400, 500])
voltages = np.array([1.5, 1.0, 0.75, 0.6, 0.5])

def rc_circuit(f, R, C):
    return 1 / (2 * np.pi * f * R * C)

popt, pcov = curve_fit(rc_circuit, frequencies, voltages)

C = popt[1]

print("Capacitance:", C)
-----

Capacitance: 1.18393 (to 5dp)
```

The errors that have arisen in this experiment may have been due to:
Cable and connection losses: The cables and connectors used in the circuit setup can have inherent losses or impedance mismatches, especially at higher frequencies. These losses can affect the voltage and current measurements and introduce errors in the calculated results.

Environmental Interference: External electromagnetic interference from nearby devices, power lines, or other sources can introduce noise and disturbances in the measured signals, leading to errors in the analysis and calculations.

Assumptions and Simplifications: The equations and models used in the analysis of AC circuits often involve simplifying assumptions. If these

assumptions are not fully valid or if there are additional factors that were not considered, it can introduce errors in the calculated results.

Modelling the Hydrogen Atom – 2/12/2022

Wednesday, May 10, 2023 4:28 PM

Exercise 1:

Calculate the overlap and Hamiltonian Matrices.

Code:

```
-----
#Exercise 1
import numpy as np
import scipy.linalg as la

n = 4
N = 4

S = np.zeros((n,n))
T = np.zeros((n,n))
A = np.zeros((n,n))

alpha = 0.15
beta = 1.5

zeta = np.array([alpha * beta ** i for i in range(N)])

for i in range(N):
    for j in range(N):
        S[i, j] = (np.pi / (zeta[i] + zeta[j])) ** (3/2)
        T[i, j] = 3 * ((zeta[i] * zeta[j] * (np.pi)**(3/3)) / (zeta[i] + zeta[j])** (5/2))
        A[i, j] = -2 * (np.pi) / (zeta[i] + zeta[j])

H = T + A
-----
```

Both the Hamiltonian and Overlap matrices are symmetrical along the diagonal showing they are correct.

Exercise 2:

Solving the GEP problem, $HC = \lambda SC$.

Code:

```
-----
#Exercise 2
import numpy as np
import scipy.linalg as la

n = 4
N = 4

S = np.zeros((n,n))
T = np.zeros((n,n))
A = np.zeros((n,n))

alpha = 0.15
```

beta = 1.5

zeta = np.array([alpha * beta ** i for i in range(N)])

for i in range(N):

for j in range(N):

S[i, j] = (np.pi / (zeta[i] + zeta[j])) ** (3/2)

T[i, j] = 3 * ((zeta[i] * zeta[j] * (np.pi)**(3/3)) / (zeta[i] + zeta[j])** (5/2))

A[i, j] = -2 * (np.pi) / (zeta[i] + zeta[j])

H = T + A

eigenvalues, eigenvectors = la.eigh(H, S)

energy = eigenvalues[0]*27.2114

print("Energy of the 1s state of hydrogen: {:.6f} eV".format(energy))

The energy of the 1s state of hydrogen was calculated to be -20.357457eV. The negative energy is correct with comparison to the tabulated energy of the 1s hydrogen energy. However, it is too large the listed energy is -13.6eV whereas the one calculated is -20.357457eV. This may be due to the calculation and assumptions made in the simulation. The size of the integration grid and type of integration could of all had an effect on the outcome.

Exercise 3:

Increasing the number of basis sets.

Code:

#exersise 3

import numpy as np

import scipy.linalg as la

n = 4

N = 4

S = np.zeros((n,n))

T = np.zeros((n,n))

A = np.zeros((n,n))

alpha = 0.15

beta = 1.5

zeta = np.array([alpha * beta ** i for i in range(N)])

for i in range(N):

for j in range(N):

S[i, j] = (np.pi / (zeta[i] + zeta[j])) ** (3/2)

T[i, j] = 3 * ((zeta[i] * zeta[j] * (np.pi)**(3/3)) / (zeta[i] + zeta[j])** (5/2))

A[i, j] = -2 * (np.pi) / (zeta[i] + zeta[j])

H = T + A

```
def ground_state_energy(n):
    eigenvalues, _ = la.eigh(H, S)
    return eigenvalues[0]*27.2114
```

```
for n in range(1, 20):
    energy = ground_state_energy(n)
    print(f"n = {n}: ground state energy: {energy:.6f} eV")
```

It returned:

```
n = 1:Energy: -20.357457 eV
n = 2:Energy: -20.357457 eV
n = 3:Energy: -20.357457 eV
n = 4:Energy: -20.357457 eV
n = 5:Energy: -20.357457 eV
n = 6:Energy: -20.357457 eV
n = 7:Energy: -20.357457 eV
n = 8:Energy: -20.357457 eV
n = 9:Energy: -20.357457 eV
n = 10:Energy: -20.357457 eV
n = 11:Energy: -20.357457 eV
n = 12:Energy: -20.357457 eV
n = 13:Energy: -20.357457 eV
n = 14:Energy: -20.357457 eV
n = 15:Energy: -20.357457 eV
n = 16:Energy: -20.357457 eV
n = 17:Energy: -20.357457 eV
n = 18:Energy: -20.357457 eV
n = 19:Energy: -20.357457 eV
```

Which is not expected. The energies should change depending on n.

Exercise 4:

Energy convergence.

Code:

#exercise 4

import numpy as np

import scipy.linalg as la

import matplotlib.pyplot as plt

n = 4

N = 4

S = np.zeros((n,n))

T = np.zeros((n,n))

A = np.zeros((n,n))

alpha = 0.15

beta = 1.5

zeta = np.array([alpha * beta ** i for i in range(N)])

for i in range(N):

for j in range(N):

S[i, j] = (np.pi / (zeta[i] + zeta[j])) ** (3/2)

$$T[i, j] = 3 * ((zeta[i] * zeta[j] * (np.pi)**(3/3)) / (zeta[i] + zeta[j])** (5/2))$$

$$A[i, j] = -2 * (np.pi) / (zeta[i] + zeta[j])$$

$$H = T + A$$

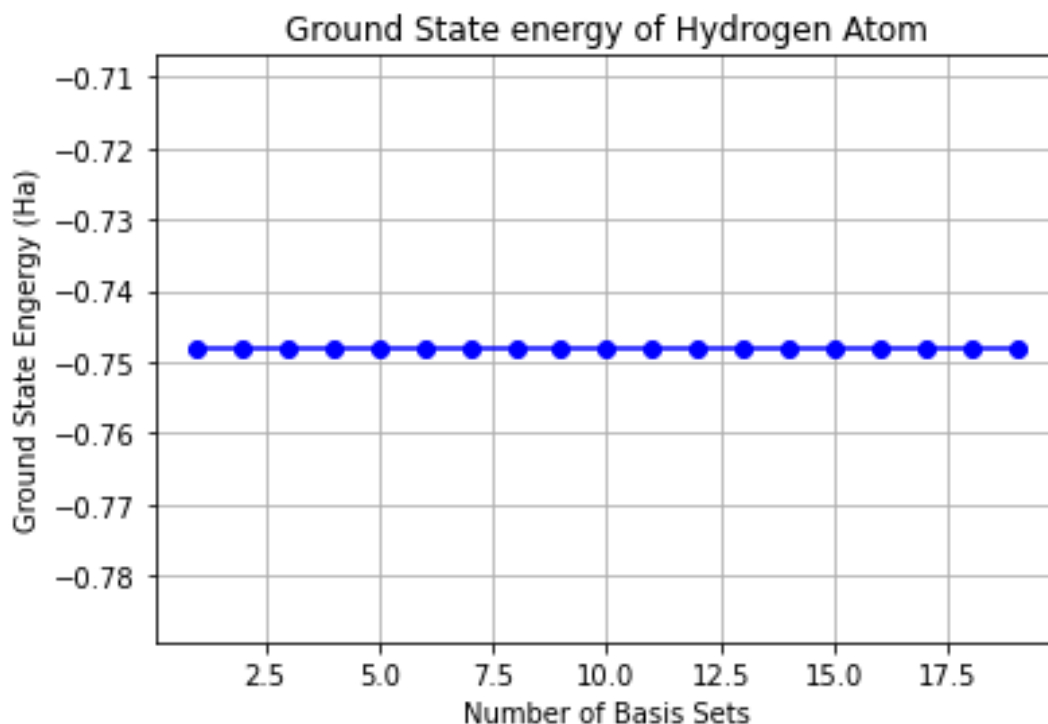
```
def ground_state_energy(n):
    eigenvalues, _ = la.eigh(H, S)
    return eigenvalues[0]
```

```
energies = []
basis_sets = []
```

```
for n in range(1, 20):
    energy = ground_state_energy(n)
    energies.append(energy)
    basis_sets.append(n)
```

```
plt.plot(basis_sets, energies, 'bo-')
plt.xlabel('Number of Basis Sets')
plt.ylabel('Ground State Energy (Ha)')
plt.title('Ground State energy of Hydrogen Atom')
plt.grid(True)
plt.show()
```

Plot of energy with number of basis sets:



The issue is that in exercise three there was only one value produced so when it came to plotting a straight line was produced. The expected plot would have been a graph converging to zero.

Next time I would research and look at other code to learn from and apply to my own experiment to help where my code went wrong.

Overall because it was a simulation experiment, uncertainty was a minimum but not zero, rounding and methods of calculation all play a part

in how the numerical values are produced. There were also model assumptions. And whenever you are simulating nature assumptions do not always hold. I would aim to calibrate and validate models in the future. As well as conduct error propagation analysis. And get code verification and validation from an expert in the field. Collaboration with other scientists would also help to provide other perspectives and insights.

Semiconductor-metal junctions – 14/3/2023

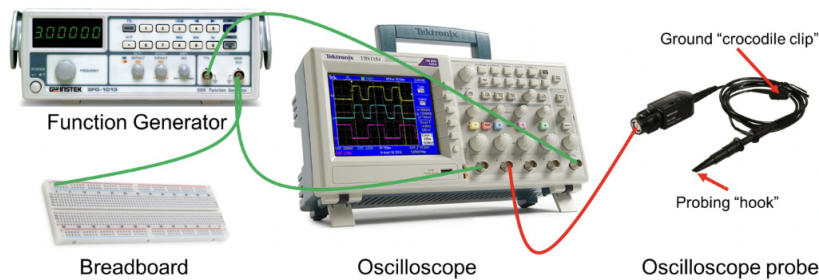
Thursday, May 11, 2023 6:57 PM

Studying the phenomenon of band bending and its effect on charges near the junction between a semiconductor and a metal

Setup:

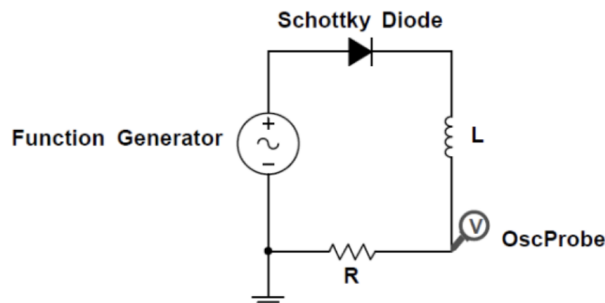
1. Use a BNC cable to connect the 'TTL' port on the function generator to the 'Ext Trig' port in the oscilloscope, and in the oscilloscope 'Trigger Menu', set the source to 'EXT'.
2. Connect the T-connector to the wave-generator output port and plug in the BNC adapter on one side and connect the RF power cable to this port.
3. Use a BNC cable to connect the free port on the T-connector to 'Channel 1' on the oscilloscope. Hit the '1' button on the oscilloscope and set the attenuation to 'x1': Probe → attenuation → x1.
4. On the wave generator, select 'sine wave' and turn on the output.
5. Press the blue 'Shift' button and immediately press the 'TTL' button to turn on the TTL output.
6. Measure the resulting waveform on the channel 1 of the oscilloscope by pushing the 'Measure' button and set the measurements to Channel 1 'mean' and Channel 1 'Pk to Pk'.
7. Tweak the amplitude and offset controls to set a waveform with 0.15 V amplitude (0.3 V peak to peak) and -1 V offset at 50 kHz.

133



Task 1:

Vary the frequency on the wave generator and find the approximate resonance frequency of the circuit below and note this value down with its uncertainty. Resonance should occur at an intermediate frequency between 20 Hz and 80 kHz

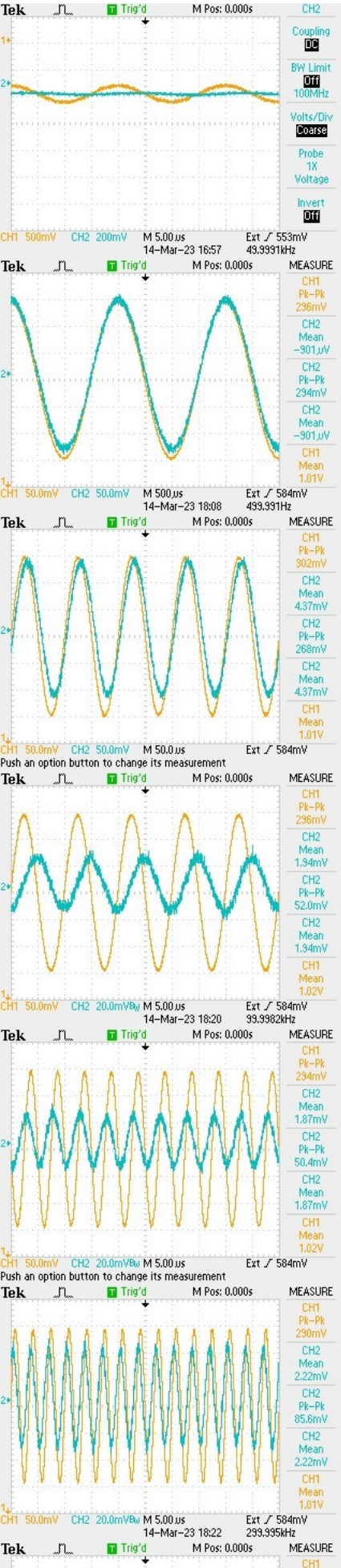


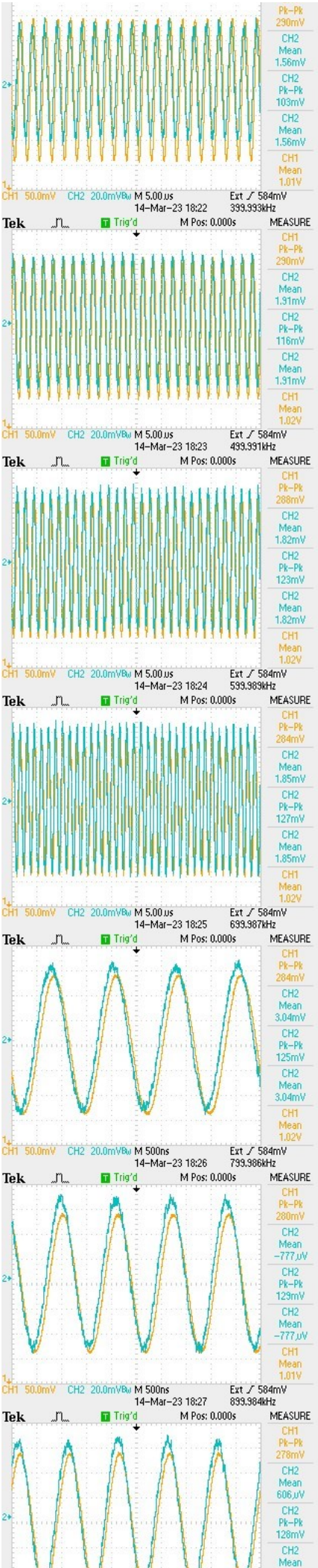
Task 2:

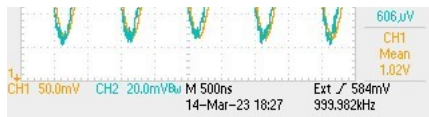
Note down the amplitude of the voltage drop across the resistor at the resonant frequency with its uncertainty.

Task 3:

Repeat the previous task for a range of frequencies from 1 kHz to 1MHz. Aiming to acquire 10 points before the resonance frequency, and 10 after.
Screen saves of oscilloscope screens at each point within 1kHz to 1MHz:







Analysis task 2:

Derivation of an expression for the voltage drop across a resistor in an RLC circuit as a function of frequency:

$$Z = r + j(X_L - X_C)$$

where: Z is the impedance of the circuit, R is the resistance, X_L is the inductive reactance. X_C is the capacitive reactance, j is the imaginary unit and f is the frequency of the circuit.

$$V_R = IR$$

where: V_R is the voltage drop across the resistor and I is the current.

$$V = IZ$$

$$I = \frac{V}{Z}$$

$$I = \frac{V}{R + j(X_L - X_C)}$$

$$V_R = \left(\frac{V}{R + j(X_L - X_C)} \right) R$$

$$V_R = \frac{VR}{\sqrt{(R^2 + (X_L - X_C)^2)}}$$

$$\text{where: } X_L = 2\pi fL \text{ and } X_C = \frac{1}{2\pi fC}$$